# Master's Thesis

## Conformal Selective Prediction with Cost-Aware Deferral Frameworks for Predictive Maintenance

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**Abstract**

Predictive maintenance systems increasingly employ machine learning models to anticipate equipment failures and support maintenance decision-making [1]. Although modern classifiers can achieve strong predictive performance, their predictions often lack reliable uncertainty estimates, making it difficult to determine when automated decisions should be trusted [2]. This limitation is particularly important in highly imbalanced and safety-critical maintenance settings, where missed failures may result in costly downtime, equipment damage, and safety risks [3]. This study investigates the integration of calibrated probabilistic modelling, conformal prediction, and cost-aware selective prediction within a unified decision-support framework for predictive maintenance. Random Forest, RUSBoost, and XGBoost models were trained on the AI4I 2020 Predictive Maintenance dataset and calibrated using isotonic regression. Split conformal prediction was employed to generate prediction sets with finite-sample coverage guarantees, while cost-aware deferral policies were used to selectively defer uncertain predictions for further inspection. In addition, a conformal-aware deferral strategy was integrated and compared against conventional confidence-based deferral approaches. Experimental results demonstrate that probability calibration improves confidence reliability, conformal prediction provides transparent uncertainty quantification, and selective deferral reduces decision risk on retained predictions. The proposed conformal-aware deferral framework achieves improved risk–coverage and cost–coverage trade-offs while maintaining strong predictive performance across multiple machine learning models. These findings suggest that, within the controlled scope of this dataset simulation, integrating uncertainty quantification with cost-sensitive decision-making may foster more trustworthy and operationally safer predictive maintenance systems.

# Contents

**Acknowledgements** **1**

**1 Introduction** **1**

1.1 Research Background . . . . . 1
1.2 Problem Statement . . . . . 2
1.3 Research Aims and Objectives . . . . . 3
1.4 Research Contributions . . . . . 4
1.5 Thesis Organization . . . . . 4

**2 Literature Review** **6**

2.1 Background . . . . . 6

**3 Proposed Methodology** **12**

3.1 Introduction . . . . . 12
3.2 Overview of the Proposed CSP-CAD Framework . . . . . 12
3.3 Dataset Description and Problem Definition . . . . . 14
3.4 Problem Definition . . . . . 16
3.5 Data Preprocessing . . . . . 16
3.6 Validation Strategy . . . . . 17
3.7 Machine Learning Models . . . . . 18
3.8 Probability Calibration using Isotonic Regression . . . . . 19
3.9 Split Conformal Prediction . . . . . 20
3.10 Conformal Prediction Procedure . . . . . 21
3.11 Confidence-Based Cost-Aware Deferral . . . . . 22
3.12 Conformal Selective Prediction with Cost-Aware Deferral (CSP-CAD) Framework . . . . . 23

**4 Analysis and Results** **28**

4.1 Baseline Classifier Evaluation . . . . . 28
4.2 Repeated Stratified CV: AUROC Comparison . . . . . 28
4.3 Repeated Stratified CV: AUPRC Comparison . . . . . 29

4.4 Comparative Analysis of Baseline Models . . . . . 30
4.5 Statistical Significance Analysis of Model Performance . . . . . 31
4.5.1 Friedman Omnibus Test . . . . . 32
4.5.2 Pairwise Wilcoxon Signed-Rank Tests . . . . . 33
4.5.3 Statistical Analysis Conclusion . . . . . 36
4.5.4 Calibration Performance Analysis . . . . . 36
4.6 Conformal Prediction Analysis . . . . . 37
4.6.1 Conformal Coverage Performance . . . . . 37
4.7 Cost-Aware Deferral Analysis . . . . . 38
4.7.1 Operational Cost Analysis . . . . . 39
4.8 Proposed Conformal-Aware Deferral Analysis . . . . . 39

**5 Conclusion and Results Discussion** **42**
5.1 Summary of Research Findings . . . . . 42
5.2 Achievement of Research Objectives . . . . . 43
5.3 Research Contributions . . . . . 44
5.4 Limitations of the Research . . . . . 45
5.5 Recommendations and Future Work . . . . . 46
5.6 Conclusion . . . . . 48
5.7 Key Concepts and Terminology . . . . . 49

# List of Figures


3.1 Conformal Selective Prediction with Cost-Aware Deferral (CSP-CAD) Framework . . . 13
3.2 Confidence-Based Cost-Aware Deferral Framework . . . 22

4.1 AUROC Results: These demonstrate substantial differences in discriminative ability among the evaluated models. . . . 29
4.2 AUPRC Comparison of Repeated Stratified Results . . . 30
4.3 Pairwise Wilcoxon Signed-Ranked Test of RUSBoost vs XGBoost . . . 33
4.4 Pairwise Wilcoxon Signed-Ranked Test of RUSBoost vs Random Forest . 34
4.5 Pairwise Wilcoxon Signed-Ranked Test of Random Forest vs XGBoost . 35
4.6 Plot of Mean Expected Calibration Error(ECE) Across 50 Runs . . . 36
4.7 Plot of Mean Conformal Coverage Across 50 Runs . . . 38
4.8 Plot of Mean Cost per Sample Across 50 Runs . . . 38
4.9 Plot of Mean Mean Risk-Coverage Curve Across 50 Runs . . . 39
4.10 Plot of Mean Risk-Coverage Curve Across 50 Runs . . . 40
4.11 Plot of Mean Error vs. Deferral Curve Across 50 Runs . . . 41

# List of Tables


3.1 Dataset Summary . . . . . 15
3.2 Input Features . . . . . 15
3.3 Hyperparameter Optimisation Strategy . . . . . 19
3.4 Prediction set summary . . . . . 22
3.5 Decision Interpretation . . . . . 23
3.6 Decision Outcomes . . . . . 24

4.1 Comparative Analysis of Baseline Models . . . . . 30

# List of Algorithms

1 Conformal Selective Prediction with Cost-Aware Deferral (CSP-CAD) Framework . . . . . 14

# Chapter 1

# Introduction

## 1.1 Research Background

The increasing adoption of Industry 4.0 technologies has transformed modern manufacturing by enabling continuous monitoring of industrial equipment through interconnected sensors and intelligent control systems. This digital transformation has generated large volumes of operational data that can be analysed to improve equipment reliability, optimise maintenance schedules, and reduce production downtime [4]. Among these applications, predictive maintenance has emerged as one of the most important data-driven maintenance strategies because it enables maintenance activities to be scheduled according to the actual condition of equipment rather than predetermined intervals or reactive failure events.

Unlike corrective maintenance, which responds after equipment failure, and preventive maintenance, which follows fixed maintenance schedules regardless of equipment condition, predictive maintenance seeks to identify signs of impending failure before breakdown occurs [5]. This enables maintenance interventions to be performed only when necessary, reducing unnecessary servicing, extending equipment lifespan, and minimising unplanned production interruptions. Consequently, predictive maintenance has become a critical component of intelligent manufacturing systems across industries such as aerospace, automotive manufacturing, energy production, and offshore engineering [6].

Recent advances in machine learning have significantly improved the ability to detect machine failures from sensor measurements. Ensemble learning algorithms, including Random Forest, Extreme Gradient Boosting (XGBoost), and RUSBoost, have demonstrated strong predictive performance by learning complex relationships between operational variables and equipment failures [7]. Despite these advances, predictive performance alone is insufficient for many industrial applications. In safety-critical environments, maintenance decisions based solely on classification accuracy may still result in costly false alarms, missed failures, or unreliable automated recommendations when pre-

diction confidence is low.

An important limitation of conventional machine learning models is that they typically produce a prediction for every observation, regardless of the model's confidence. While many algorithms output probabilities, these probabilities are often poorly calibrated and may not accurately reflect the true likelihood of failure [8]. As a result, highly confident predictions can still be incorrect, reducing trust in automated maintenance systems and increasing operational risk.

Reliable uncertainty estimation has therefore become an increasingly important research area within trustworthy artificial intelligence. Instead of relying exclusively on point predictions, uncertainty-aware machine learning seeks to quantify the confidence associated with each prediction so that uncertain cases can be identified before maintenance decisions are made. Probability calibration improves the reliability of predicted probabilities, while conformal prediction provides statistically valid prediction sets with finite-sample marginal coverage guarantees under the exchangeability assumption [9]. Together, these techniques enable machine learning systems to communicate not only what they predict but also how confident they are in those predictions.

Selective prediction extends this concept by allowing an intelligent system to abstain from making automated decisions when uncertainty exceeds an acceptable level [10]. Rather than forcing a potentially incorrect prediction, uncertain observations may instead be referred for further inspection or expert review. Such decision-support mechanisms are particularly attractive in predictive maintenance, where the consequences of incorrect decisions often differ substantially. Missing an impending machine failure may lead to catastrophic production losses, whereas unnecessary maintenance primarily incurs inspection costs and equipment downtime. Consequently, maintenance decision-making should account not only for predictive accuracy but also for the operational costs associated with different types of decision errors.

Motivated by these challenges, this research develops a Conformal Selective Prediction with Cost-Aware Deferral (CSP-CAD) Framework for predictive maintenance. The proposed framework integrates calibrated probabilistic machine learning, Split Conformal Prediction, and cost-aware selective prediction within a unified decision-support pipeline. By allowing uncertain predictions to be deferred while incorporating operational cost considerations into decision-making, the framework aims to improve the reliability, transparency, and practical usefulness of machine learning-assisted predictive maintenance.

## 1.2 Problem Statement

Machine learning has significantly improved predictive maintenance by enabling automated identification of machine failures from operational sensor data. However, most predictive maintenance systems continue to prioritise predictive performance while assum-

ing that every observation should receive an automated prediction [11]. In safety-critical industrial environments, this assumption may lead to unreliable maintenance decisions when models encounter uncertain operating conditions.

Three important challenges remain. First, probability estimates produced by many machine learning models are often poorly calibrated, reducing the reliability of confidence estimates used during decision-making. Second, conventional classifiers rarely provide formal measures of predictive uncertainty, making it difficult to distinguish reliable predictions from uncertain ones. Third, most predictive maintenance frameworks do not explicitly incorporate operational costs when deciding whether to issue an automated prediction or defer the decision for further inspection.

Although probability calibration, conformal prediction, and selective prediction have each been studied independently, relatively few predictive maintenance studies have integrated these techniques within a unified decision-support framework that simultaneously addresses predictive uncertainty and operational decision costs. This limitation motivates the present research.

The central research problem addressed in this thesis is therefore: **How can calibrated probabilistic machine learning, Split Conformal Prediction, and cost-aware selective prediction be integrated to improve the reliability and operational effectiveness of predictive maintenance under severe class imbalance?**

## 1.3 Research Aims and Objectives

The aim of this research is to develop and evaluate a Conformal Selective Prediction with Cost-Aware Deferral (CSP-CAD) framework that integrates calibrated machine learning, conformal prediction, and cost-aware selective prediction to improve uncertainty-aware decision support for predictive maintenance.

To achieve this aim, the following objectives were established:

1. Develop predictive machine learning models capable of identifying machine failures using the AI4I 2020 predictive maintenance dataset.

2. Improve the reliability of predicted probabilities through isotonic regression probability calibration.

3. Quantify predictive uncertainty using Split Conformal Prediction with probability-based nonconformity scores.

4. Develop a cost-aware selective prediction framework that balances false-negative, false-positive, and deferral costs when making maintenance decisions.

5. Integrate conformal prediction information into the selective prediction framework to support uncertainty-aware deferral decisions.

6. Evaluate the proposed framework using repeated stratified cross-validation and operational performance measures including calibration quality, conformal coverage, Risk-Coverage, Cost-Coverage, Error-vs-Deferral, and Cost per Sample analyses

## 1.4 Research Contributions

This thesis makes the following principal contributions to uncertainty-aware predictive maintenance.

1. **Integrated Predictive Maintenance Framework**

A unified Conformal Selective Prediction with Cost-Aware Deferral (CSP-CAD) framework is developed by integrating calibrated probabilistic machine learning, Split Conformal Prediction, and selective prediction into a single predictive maintenance pipeline.

2. **Cost-Aware Deferral Strategy**

A selective prediction policy is implemented that explicitly considers the operational costs associated with false negatives, false positives, and deferred decisions, enabling maintenance recommendations to be optimised according to application-specific priorities.

3. **Adaptation of Conformal-Aware Deferral to Predictive Maintenance**

The research adapts a conformal-aware deferral strategy to the predictive maintenance domain by combining confidence-based decision thresholds with conformal prediction sets to identify uncertain observations requiring further inspection.

4. **Comprehensive Experimental Evaluation**

The proposed framework is evaluated using repeated stratified cross-validation consisting of 5 folds and 10 repeats (50 evaluation runs). Performance is assessed using conventional classification metrics together with calibration quality, conformal validity, and operational decision-support metrics, providing a robust assessment of both predictive performance and uncertainty-aware decision making.

## 1.5 Thesis Organization

This research thesis is structured into five main chapters to systematically address the research aim and objectives:

- **Chapter 1**

Introduction: This chapter provides an overview of the research topic, outlines the research's aim and specific objectives, details the motivation and justification for the study, and presents the overall arrangement of the thesis.

- **Chapter 2**

Literature Review: This chapter offers an in-depth analysis of existing prognostic methodologies, critically examining the evolution of predictive maintenance from standard point-estimation towards robust, uncertainty-aware decision support systems.

- **Chapter 3**

Methodology: This chapter presents the proposed Conformal Selective Prediction with Cost-Aware Deferral Framework, including model training, calibration procedures, conformal prediction methodology, and deferral optimisation. It details the rigorous mathematical formulation of the risk-cost utility function, establishing the thresholding mechanism required to optimise system-level decision-making.

- **Chapter 4**

Analysis and Results Discussion: This chapter presents the experimental results and comparative analyses across predictive performance, calibration quality, conformal prediction, and deferral strategies.

- **Chapter 5**

Conclusion and Recommendation Discussion: This concluding chapter synthesises the key empirical findings, summarises the framework's overall contribution to cost-efficient maintenance strategies, and outlines potential avenues for future research in industrial prognostic systems. This chapter also discusses the implications of the findings, limitations of the study, and opportunities for future research.

# Chapter 2

# Literature Review

The review aims to establish the current state of knowledge, identify key methodologies, and highlight research gaps that this thesis seeks to address.

## 2.1 Background

This section introduces the fundamental concepts and theoretical frameworks that underpin the methodology and objectives of this research.

### 2.1.1 Theoretical Foundations of Predictive Maintenance

Predictive maintenance (PdM) is a condition-based maintenance strategy that aims to anticipate equipment failures using operational data collected from industrial systems [12]. Unlike corrective maintenance, which responds after failures occur, and preventive maintenance, which follows fixed maintenance schedules, predictive maintenance estimates equipment health to enable maintenance activities before failures occur. This approach improves equipment reliability, reduces unplanned downtime, and lowers maintenance costs.

The rapid adoption of Industry 4.0 technologies has accelerated the use of predictive maintenance through the integration of sensors, Industrial Internet of Things (IIoT) devices, and data-driven monitoring systems. These technologies continuously collect operational information, allowing machine learning algorithms to identify patterns associated with equipment degradation and impending failures. Consequently, predictive maintenance has become widely adopted across manufacturing, transportation, energy, and other safety-critical industries.

Despite these advances, predictive maintenance remains challenging because industrial datasets are typically highly imbalanced, with failure events occurring far less frequently than normal operating conditions. In addition, many existing approaches prioritise predictive accuracy while providing limited information about the reliability or uncertainty

of their predictions [13]. These limitations reduce confidence in automated maintenance decisions, particularly in high-risk operational environments.

Addressing these challenges has motivated the increasing adoption of machine learning techniques capable of learning complex failure patterns from industrial data. The following section reviews the application of machine learning methods for predictive maintenance and discusses their strengths and limitations.

### 2.1.2 Machine Learning for Predictive Maintenance

Machine learning has become a key enabling technology for predictive maintenance by learning patterns from historical sensor measurements to predict equipment failures before they occur. Unlike traditional statistical or rule-based approaches, machine learning algorithms can model complex and non-linear relationships between operational variables, making them well suited to modern industrial systems where large volumes of sensor data are continuously generated [14].

A wide range of supervised learning algorithms has been applied to predictive maintenance, including Logistic Regression, Decision Trees, Support Vector Machines, Random Forest, Gradient Boosting, and Extreme Gradient Boosting (XGBoost). Ensemble learning methods have received particular attention because they combine multiple weak or diverse learners to improve predictive accuracy, robustness, and generalisation [15]. Studies have shown that algorithms such as Random Forest and XGBoost consistently achieve strong performance in machine failure prediction due to their ability to capture complex interactions among operating variables while reducing overfitting. Similarly, RUSBoost combines random undersampling with boosting to improve minority class detection in highly imbalanced datasets.

Despite these advances, the performance of machine learning models cannot be evaluated solely by conventional classification metrics such as accuracy or the Area Under the Receiver Operating Characteristic Curve (AUROC). In predictive maintenance, failure events are typically rare, causing standard classifiers to become biased towards the majority class [16]. As a result, models with high overall accuracy may still perform poorly when identifying the critical minority failure cases.

These challenges highlight the importance of developing machine learning models that not only achieve strong predictive performance but also effectively handle class imbalance and produce reliable predictions. The following section therefore reviews existing approaches for addressing class imbalance in predictive maintenance applications.

### 2.1.3 Machine Learning under Class Imbalance

Class imbalance is one of the most significant challenges in predictive maintenance because machine failures occur much less frequently than normal operating conditions. As a result,

machine learning models trained on imbalanced datasets tend to favour the majority class, often producing high overall accuracy while failing to correctly identify the relatively few failure events that are of greatest operational importance. This limitation makes conventional accuracy an inadequate measure of model performance in safety-critical applications.

To address this challenge, various imbalance learning strategies have been proposed. Data-level approaches, such as random oversampling, random undersampling, and the Synthetic Minority Over-sampling Technique (SMOTE), attempt to rebalance the class distribution before model training [17]. Algorithm-level approaches, including cost-sensitive learning and ensemble methods such as RUSBoost, instead modify the learning process to improve the detection of minority class instances without relying solely on data resampling [18].

Although these techniques have demonstrated improvements in minority class prediction, challenges remain. Many methods focus primarily on increasing recall or reducing class imbalance but provide limited information regarding the confidence or uncertainty associated with their predictions. In predictive maintenance, where false negatives may lead to unexpected equipment failures and false positives may result in unnecessary maintenance, reliable uncertainty estimation is equally important for supporting informed maintenance decisions.

These limitations have motivated research into uncertainty-aware machine learning methods that complement predictive performance with reliable confidence estimates. The following section reviews probability calibration techniques that improve the reliability of predicted probabilities and provide the foundation for subsequent uncertainty quantification methods.

### 2.1.4 Probability Calibration

Probability calibration is the process of adjusting the probability estimates produced by a machine learning model so that predicted confidence levels more accurately reflect observed event frequencies [19]. While many classifiers achieve high predictive accuracy, their probability estimates are often overconfident or underconfident, making them unreliable for decision-making in safety-critical applications. In predictive maintenance, poorly calibrated probabilities may lead to inappropriate maintenance actions by either underestimating the likelihood of machine failure or generating unnecessary interventions.

Several post-hoc calibration techniques have been proposed to improve the reliability of probabilistic predictions. Common approaches include Platt Scaling, Isotonic Regression, and Temperature Scaling, each aiming to transform raw model outputs into better-calibrated probability estimates. Among these methods, Isotonic Regression has been widely adopted for ensemble learning models because it provides a flexible, non-

parametric calibration function without assuming a predefined probability distribution [19].

Although probability calibration improves the reliability of confidence estimates, calibrated probabilities alone do not quantify the uncertainty associated with individual predictions. Consequently, additional uncertainty quantification methods are required to determine when machine learning predictions can be trusted and when further inspection may be necessary. This motivates the use of uncertainty-aware learning techniques, which are discussed in the following section.

### 2.1.5 Uncertainty Quantification

Uncertainty quantification (UQ) aims to measure the reliability of machine learning predictions by estimating the degree of confidence associated with each decision. Unlike conventional classifiers that provide only a predicted class label, uncertainty-aware models communicate how certain they are about their predictions [20]. This capability is particularly important in safety-critical applications such as predictive maintenance, where incorrect decisions may result in unexpected equipment failures, increased operational costs, or unnecessary maintenance interventions.

In machine learning, uncertainty is commonly categorised as **aleatoric uncertainty**, which arises from inherent variability or noise in the data, and **epistemic uncertainty**, which reflects uncertainty in the model due to limited or incomplete knowledge [21]. Various approaches have been proposed to quantify predictive uncertainty, including Bayesian methods, ensemble learning, Monte Carlo sampling techniques, and distribution-free methods such as conformal prediction. Each approach offers different trade-offs between computational complexity, interpretability, and statistical guarantees.

Although these methods improve the transparency of machine learning predictions, uncertainty estimates must also be translated into practical decision support. This has motivated increasing interest in conformal prediction, which provides statistically valid prediction sets with finite-sample marginal coverage guarantees under the exchangeability assumption. The following section reviews conformal prediction and its application to uncertainty-aware predictive maintenance.

### 2.1.6 Conformal Prediction

Conformal prediction (CP) is a distribution-free uncertainty quantification framework that complements machine learning models by producing prediction sets with finite-sample marginal coverage guarantees under the assumption of exchangeability [9]. Rather than assigning a single class label, conformal prediction returns a set of plausible labels, allowing the uncertainty associated with each prediction to be explicitly represented. This characteristic makes conformal prediction particularly attractive for safety-critical appli-

cations where understanding prediction reliability is as important as predictive accuracy.

Several variants of conformal prediction have been proposed, including Full Conformal Prediction, Split Conformal Prediction, Mondrian Conformal Prediction, and Adaptive Conformal Prediction. Among these approaches, Split Conformal Prediction has gained widespread adoption because it is computationally efficient, model-agnostic, and easily integrated with existing machine learning algorithms [9]. Recent studies have successfully applied conformal prediction to medical diagnosis, autonomous systems, and industrial fault detection, demonstrating its ability to produce statistically valid uncertainty estimates while maintaining strong predictive performance.

Despite these advances, the application of conformal prediction in predictive maintenance remains relatively limited. Existing studies have primarily focused on improving uncertainty estimation or prediction coverage, while less attention has been given to integrating conformal prediction with operational decision-making processes. In particular, relatively few studies have investigated how conformal prediction can support selective prediction and cost-aware maintenance decisions by identifying observations that should be deferred for additional inspection.

These limitations have motivated growing interest in combining conformal prediction with selective prediction frameworks that explicitly incorporate uncertainty into automated decision-making. The following section reviews selective prediction and cost-aware deferral strategies that enable machine learning systems to abstain from uncertain predictions while balancing operational risks and maintenance costs.

### 2.1.7 Selective Prediction and Cost-Aware Deferral

Selective prediction extends conventional machine learning by allowing a model to abstain from making predictions when uncertainty exceeds an acceptable level [22]. Instead of forcing an automated decision for every observation, selective prediction identifies uncertain cases and refers them for additional analysis or expert review. This approach is particularly valuable in safety-critical applications, where the consequences of incorrect predictions may outweigh the benefits of full automation.

Several selective prediction strategies have been proposed in the literature, including confidence-based rejection, reject-option classification, and human-in-the-loop decision support. These methods typically rely on prediction confidence or uncertainty estimates to determine whether an automated prediction should be accepted or deferred. By abstaining from uncertain predictions, selective prediction can improve the reliability of automated systems while reducing the risk of costly decision errors [23].

More recently, cost-aware deferral frameworks have been introduced to explicitly incorporate the asymmetric costs associated with different decision outcomes. Rather than maximising predictive accuracy alone, these approaches optimise decision-making by

balancing the costs of false negatives, false positives, and deferred decisions according to application-specific requirements [24]. In predictive maintenance, such cost-sensitive strategies are particularly important because undetected equipment failures may result in substantially greater operational losses than unnecessary maintenance inspections.

Although selective prediction and cost-aware deferral have demonstrated promising results across several application domains, their integration with calibrated probabilities and conformal prediction remains relatively unexplored in predictive maintenance. This limitation motivates the research presented in this thesis, which integrates these complementary techniques into a unified uncertainty-aware maintenance decision-support framework.

### 2.1.8 Research Gap and Summary

Existing work has shown that machine learning can accurately predict equipment failures, while probability calibration improves confidence estimation, conformal prediction provides statistically valid uncertainty estimates, and selective prediction reduces prediction risk by abstaining on uncertain observations.

However, these approaches have largely been investigated independently.

Few predictive maintenance studies integrate calibrated probabilistic machine learning, conformal prediction, and cost-aware selective prediction within a unified decision-support framework that explicitly incorporates operational costs when deciding whether uncertain predictions should be automated or deferred [25].

Furthermore, many predictive maintenance studies continue to evaluate proposed methods using a single train–test split, limiting the robustness and generalisability of reported performance. More comprehensive evaluation strategies are needed to assess both predictive performance and uncertainty-aware decision quality under repeated data partitions.

To address these gaps, this research adapts a **Conformal Selective Prediction with Cost-Aware Deferral (CSP-CAD) framework** that integrates calibrated probabilistic machine learning, standard Split Conformal Prediction using probability-based nonconformity scores, and cost-aware selective prediction within a unified predictive maintenance pipeline. This builds on the work of Kwon and Kim from their published report on Conformal selective prediction with cost aware deferral for safe clinical triage under distribution shift [25]. The framework is evaluated using repeated stratified cross-validation to provide a robust assessment of predictive performance, probability calibration, conformal validity, operational cost, and uncertainty-aware decision making.

# Chapter 3

# Proposed Methodology

## 3.1 Introduction

This chapter delineates the architectural design of the cost-aware conformal deferral framework, structured to address the limitations of standard prognostic models in non-stationary industrial environments. The framework utilises the AI4I 2020 Predictive Maintenance data set to systematically evaluate the efficacy of the proposed uncertainty-quantified maintenance scheduling. By leveraging conformal prediction, the methodology establishes rigorous, distribution-free reliability bounds for operation mode estimates, moving beyond traditional point-prognostic approaches that often lack explicit uncertainty quantification [26].

## 3.2 Overview of the Proposed CSP-CAD Framework

Figure 3.1 presents the overall architecture of the **Conformal Selective Prediction with Cost-Aware Deferral (CSP-CAD)** framework. The framework integrates machine learning, probability calibration, conformal prediction, and selective prediction into a unified decision-support pipeline for predictive maintenance. Rather than requiring an automated prediction for every observation, the framework identifies uncertain cases and defers them for further inspection, thereby improving decision reliability in safety-critical and highly imbalanced environments.

The framework consists of six sequential stages. First, the AI4I 2020 Predictive Maintenance dataset is preprocessed through feature encoding and standardisation. Next, model performance is evaluated using repeated stratified cross-validation, within which separate training, calibration, and testing subsets are created for each fold. Three ensemble learning algorithms—Random Forest, RUSBoost, and XGBoost—are then trained to predict machine failures from the prepared data.

Following model training, isotonic regression is applied to calibrate predicted prob-

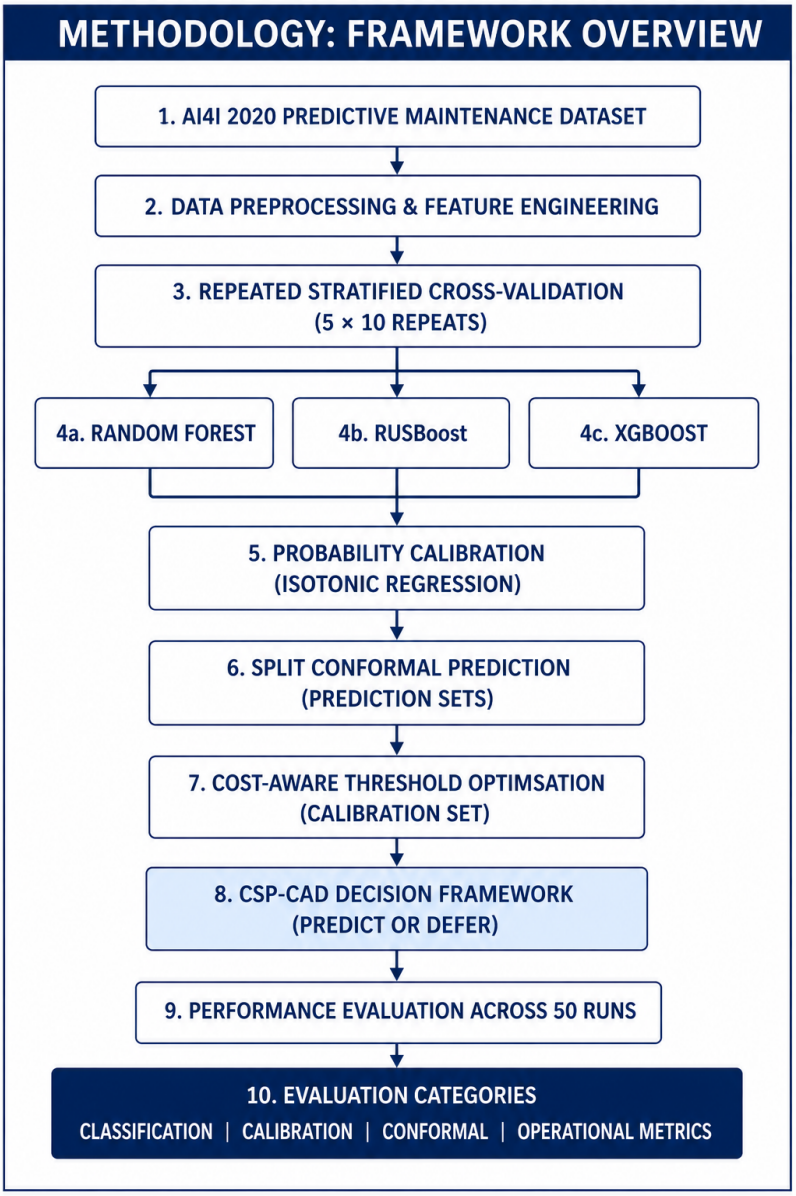



Figure 3.1: Conformal Selective Prediction with Cost-Aware Deferral (CSP-CAD) Framework

abilities. These calibrated probabilities are subsequently used to construct conformal prediction sets using Split Conformal Prediction and to optimise confidence thresholds under a cost-aware objective. The resulting calibrated probabilities and conformal prediction sets are integrated within the proposed CSP-CAD decision framework, which issues automated predictions for confident observations while deferring uncertain cases for human review.

Finally, the framework is evaluated using predictive, calibration, conformal, and operational performance metrics aggregated across fifty repeated cross-validation runs. This evaluation provides a robust assessment of predictive accuracy, uncertainty quantification, and decision quality while reducing the dependence of the results on a single data

partition.

The algorithm describing the entire process is expressed below:

**Algorithm 1:** Conformal Selective Prediction with Cost-Aware Deferral (CSP-CAD) Framework

Dataset $D$, number of folds $K = 5$, repetitions $R = 10$, significance level $\alpha$, cost parameters $(C_{FN}, C_{FP}, C_{DEF})$.
Mean classification, calibration, conformal and operational performance metrics.
**Phase 1: Data Preparation**
    Preprocess dataset → Remove identifier variables → Remove identifier variables → Encode categorical features → Standardise numerical features.
**Phase 2: Model Development**
    **for** $r = 1$ *to* $R$ **do**
        Generate $K$ stratified folds
            **for** *each fold* **do**
                Split data into: Training subset, Calibration subset, Test subset
Train machine learning classifier Random Forest, RUSBoost, XGBoost
Calibrate predicted probabilities using Isotonic Regression.
**Phase 3: Uncertainty Quantification**
Compute nonconformity scores, Estimate conformal quantile threshold, Construct Split Conformal prediction sets
**Phase 4: Decision Support**
Optimise confidence threshold by minimising expected operational cost on the calibration subset
**if** *Prediction set is singleton* ***and*** *confidence* $\geq \tau$ **then**
        Issue automated prediction **else**
        Defer observation for human inspection
**Phase 5: Performance Evaluation**
Evaluate performance on the independent test subset
    Compute: Classification metrics, Calibration metrics, Conformal prediction metrics, Operational metrics,
Aggregate results across all 50 evaluation runs
**return** Mean classification, calibration, conformal and operational performance metrics

## 3.3 Dataset Description and Problem Definition

The experiments in this study were conducted using the **AI4I 2020 Predictive Maintenance Dataset**, a publicly available benchmark dataset for machine failure prediction.

The dataset contains **10,000 observations** representing simulated industrial operating conditions, where each observation corresponds to the state of a manufacturing machine during operation. Machine failure is treated as a binary classification problem, with the target variable indicating whether a failure occurred during a given operating cycle.

The dataset comprises a combination of categorical and continuous operating variables that describe machine conditions. Product type is represented as a categorical feature, while air temperature, process temperature, rotational speed, torque, and tool wear are recorded as numerical variables. These features provide the information used by the machine learning models to estimate the probability of machine failure.

Although the AI4I dataset is generated from simulated industrial processes, it has been widely adopted as a benchmark for evaluating predictive maintenance algorithms because it captures realistic operating behaviour and severe class imbalance commonly encountered in industrial environments. This makes it suitable for evaluating uncertainty-aware predictive maintenance frameworks.

Table 3.1: Dataset Summary

| **Property** | **Value** |
|---|---|
| Dataset | AI4I 2020 Predictive Maintenance |
| Total observations | 10,000 |
| Prediction task | Binary Classification |
| Target varibale | Machine Failure |
| Failure Observations | 339 |
| Normal Observations | 9,661 |
| Failure Rate | 3.39% |
| Number of Input Features | 6 |
| Feature Types | Numerical and Categorical |

Table 3.2: Input Features

| **Feature** | **Type** | **Description** |
|---|---|---|
| Product Type | Categorical | Product quality category (L, M, H) |
| Air Temperature | Numerical | Ambient operating temperature (K) |
| Process Temperature | Numerical | Ambient operating temperature (K) |
| Rotational Speed | Numerical | Rotational speed (rpm) |
| Torque | Numerical | Applied torque (Nm) |
| Tool Wear | Numerical | Tool wear duration (minutes) |
| Machine Failure | Target | Binary failure label |

## 3.4 Problem Definition

The objective of this research is to develop an uncertainty-aware predictive maintenance framework that predicts machine failure while identifying observations for which automated predictions are insufficiently reliable. Let $x \in \mathbb{R}^d$ denote the feature vector describing a machine's operating condition and $y \in \{0, 1\}$ denote the corresponding machine state, where $y = 1$ represents machine failure and $y = 0$ represents normal operation.

$$P(y = 1 \mid x) \qquad \ldots\ldots (1)$$

The framework estimates the probability of machine failure using supervised machine learning models. These probability estimates are subsequently calibrated, used to construct conformal prediction sets, and incorporated into a cost-aware selective prediction strategy that either issues an automated maintenance recommendation or defers uncertain cases for further inspection.

## 3.5 Data Preprocessing

### 3.5.1 Overview

Before model training, the AI4I 2020 dataset was preprocessed to ensure consistency across all experiments. The preprocessing pipeline consisted of feature selection, categorical feature encoding, numerical feature standardisation, and preparation of the data for repeated stratified cross-validation. These steps were applied identically in every cross-validation iteration to ensure a fair comparison between the evaluated machine learning models.

### 3.5.2 Preprocessing Steps

The preprocessing procedure consisted of the following sequential steps:

#### Step 1: Removal of Identifier Variables

The **UDI** and **Product ID** attributes were removed because they uniquely identify observations and do not contain predictive information related to machine failure. Retaining identifier variables may introduce unnecessary noise into the learning process without improving predictive performance.

#### Step 2: Categorical Feature Encoding

The **Product Type** feature, consisting of the categories **Low (L)**, **Medium (M)**, and **High (H)**, was transformed into numerical representations using label encoding to enable

processing by the selected machine learning algorithms.

#### Step 3: Feature Standardisation

The numerical features—including air temperature, process temperature, rotational speed, torque, and tool wear—were standardised using z-score normalisation. Standardisation was fitted using the training data within each cross-validation fold and subsequently applied to the corresponding calibration and test subsets to prevent data leakage.

## 3.6 Validation Strategy

### 3.6.1 Overview

To obtain robust and reproducible estimates of model performance, this study employed **Repeated Stratified 5-Fold Cross-Validation with 10 repetitions**, resulting in **50 independent evaluation runs** for each machine learning model. This validation strategy reduces the dependence of the reported results on a single train–test split while preserving the proportion of failure and non-failure observations in every fold.

Within each cross-validation iteration, the training data were further divided into **training** and **calibration** subsets. The training subset was used to fit the machine learning model, whereas the calibration subset was reserved for probability calibration, conformal prediction, and cost-aware threshold optimisation. The test subset remained completely unseen until the final evaluation stage.

### 3.6.2 Validation Procedure

For each repetition, the dataset was first partitioned into five stratified folds while preserving the proportion of failure and non-failure observations. One fold was reserved for testing, and the remaining folds formed the development data. The development data were subsequently divided into training and calibration subsets. The training subset was used to fit the machine learning classifier, whereas the calibration subset was used exclusively for probability calibration, conformal prediction, and cost-aware threshold optimisation.

All preprocessing operations that required parameter estimation, including feature standardisation, were fitted using only the training subset before being applied to the calibration and test subsets. This procedure prevented information leakage and ensured an unbiased evaluation of predictive performance.

After model training and calibration, predictions were generated for the independent test subset. This process was repeated for all folds and repetitions, yielding fifty independent evaluation runs. The reported performance metrics correspond to the mean values obtained across these runs.

## 3.7 Machine Learning Models

### 3.7.1 Overview

Three ensemble machine learning algorithms—**Random Forest (RF)**, **RUSBoost**, and **Extreme Gradient Boosting (XGBoost)**—were selected as the primary classifiers in the proposed framework. In addition, **Logistic Regression (LR)** and **Decision Tree (DT)** were included as baseline models to provide reference performance against more advanced ensemble methods.

The selected models represent a progression from simple linear classification to tree-based ensemble learning, enabling a comprehensive comparison under the proposed CSP-CAD framework.

### 3.7.2 Logistic Regression

Logistic Regression was included as a baseline classifier because of its simplicity, interpretability, and widespread use in binary classification problems. The model estimates the probability of machine failure using a logistic function applied to a linear combination of the input features. Although its capacity to capture complex nonlinear relationships is limited, it provides an important benchmark against which more sophisticated ensemble methods can be evaluated.

### 3.7.3 Decision Tree

Decision Tree was selected as an interpretable baseline capable of modelling nonlinear relationships through recursive feature partitioning. The model constructs a hierarchy of decision rules that partition the feature space into increasingly homogeneous subsets. Its transparent decision structure provides an easily interpretable benchmark, although individual trees are generally more susceptible to overfitting than ensemble methods.

### 3.7.4 Random Forest

Random Forest was selected as one of the primary classifiers due to its robustness, ability to model nonlinear interactions, and resistance to overfitting. The algorithm combines multiple decision trees trained on bootstrap samples while introducing random feature selection at each split. Final predictions are obtained through majority voting across the ensemble.

$$\hat{y} = \text{mode}(T_1(x), T_2(x), \ldots, T_K(x)) \qquad \ldots\ldots (2)$$

### 3.7.5 RUSBoost

RUSBoost was selected because it combines random under-sampling with adaptive boosting, making it particularly suitable for highly imbalanced classification problems such as predictive maintenance. At each boosting iteration, majority-class observations are randomly under-sampled before training a weak learner. Subsequent iterations focus increasingly on previously misclassified observations, allowing the ensemble to improve minority-class detection while mitigating class imbalance.

### 3.7.6 XGBoost

XGBoost was selected because of its computational efficiency, regularisation mechanisms, and strong predictive performance on structured tabular datasets. The model builds an additive ensemble of decision trees by sequentially minimising a regularised objective function through gradient boosting. Regularisation helps reduce model complexity and improves generalisation performance.

### 3.7.7 Hyperparameter Configuration

Table 3.3: Hyperparameter Optimisation Strategy

| **Aspect** | **Description** |
|---|---|
| Configuration files | YAML configuration files |
| Hyperparameter search | Grid Search |
| Optimisation objective | Highest validation performance |
| Validation strategy | Repeated Stratified Cross-Validation |
| Final model | Best-performing configuration |

## 3.8 Probability Calibration using Isotonic Regression

The predicted probabilities produced by the machine learning classifiers were calibrated using **Isotonic Regression** before uncertainty estimation and selective prediction. Calibration was performed on the calibration subset within each repeated cross-validation iteration to improve the agreement between predicted probabilities and observed outcome frequencies.

The calibrated probabilities were subsequently used for Split Conformal Prediction and cost-aware threshold optimisation within the proposed CSP-CAD framework.

### 3.8.1 Calibration Procedure

For each cross-validation iteration, the classifier was first trained using the training subset. The trained classifier then generated probability estimates for the corresponding calibration subset.

Unlike parametric calibration methods, Isotonic Regression does not assume a predefined functional relationship between predicted probabilities and true event frequencies. Instead, it estimates a piecewise constant monotonic function that preserves the ordering of probability estimates while improving probability reliability.

The fitted calibration model was subsequently applied to the independent test subset to obtain calibrated probability estimates used throughout the remainder of the CSP-CAD framework.The calibration mapping can be expressed as

$$\hat{p}_{\text{cal}} = g(\hat{p}) \qquad \ldots\ldots (3)$$

where:

- $\hat{p}$ denotes the predicted probability produced by the classifier,
- $g()$ represents the isotonic regression mapping, and
- $\hat{p}_{\text{cal}}$ denotes the calibrated probability.

### 3.8.2 Calibration Evaluation Metrics

The calibration quality was evaluated using the following metrics:

**Expected Calibration Error (ECE):** Measures calibration reliability.

**Brier Score:** Measures probability accuracy.

**Log Loss:** Measures probabilistic prediction quality

Lower values of ECE, Brier Score, and Log Loss indicate better-calibrated probabilistic predictions. The calibrated probabilities obtained through Isotonic Regression provide reliable confidence estimates for the subsequent uncertainty quantification stage. These probabilities are used to construct conformal prediction sets and optimise selective prediction decisions within the proposed CSP-CAD framework.

## 3.9 Split Conformal Prediction

Split Conformal Prediction was employed to quantify prediction uncertainty by constructing prediction sets with a predefined confidence level. Unlike conventional classifiers that always produce a single class prediction, conformal prediction provides a set of plausible class labels while offering finite-sample coverage guarantees under the assumption of exchangeable data [9].

Within the proposed CSP-CAD framework, Split Conformal Prediction was applied after probability calibration. The calibrated probabilities were used to compute nonconformity scores on the calibration subset, estimate the conformal threshold, and generate prediction sets for unseen test observations. These prediction sets were subsequently incorporated into the selective prediction framework to distinguish between confident and uncertain predictions.

## 3.10 Conformal Prediction Procedure

For each repeated cross-validation iteration, calibrated probabilities obtained from the preceding stage were used to compute nonconformity scores on the calibration subset. The nonconformity score measures the disagreement between the predicted probability assigned to the true class and the observed label, thereby quantifying how unusual each calibration example appears relative to the trained model. The empirical distribution of these scores was then used to estimate a quantile threshold corresponding to the selected significance level, $\alpha$. For every unseen test observation, prediction sets were constructed by including all class labels whose nonconformity scores satisfied the conformity criterion. Consequently, singleton prediction sets indicate confident predictions, whereas larger prediction sets represent increased predictive uncertainty. The **nonconformity score** is represented by the equation:

$s_{(x,y)} = 1 - p(x|y) \qquad \ldots\ldots (4),$

where $p(x|y)$ represents the calibrated probability assigned to the true class label.

The **conformal quantile threshold** is defined as:

$q_{1-\alpha} = \text{Quantile}_{1-\alpha}(\{s_i\}_{i=1}^{n_{\text{cal}}}) \qquad \ldots\ldots (5),$

where $\{s_i\}_{i=1}^{n_{\text{cal}}}$ represents the set of nonconformity scores derived from the calibration dataset, and $n$ is the number of samples in that set. A confidence level of 90% ($\alpha = 0.10$) was adopted throughout this research, chosen to establish a strategic balance between predictive precision and the operational costs inherent in the deferral framework.

The prediction set is defined as: $\Gamma(x) = y \in Y : s(x, y) \leq q_{1-\alpha} \qquad \ldots\ldots (6)$

where $\Gamma(x)$ denotes the prediction set for input $x$, $Y$ is the set of possible class labels, $s(x, y)$ is the nonconformity score for class , and $q_{1-\alpha}$ is the conformal quantile threshold. The prediction set contains all class labels whose nonconformity scores do not exceed the threshold. In this study, singleton prediction sets (e.g., $\{0\}$ or $\{1\}$) indicate confident predictions, whereas prediction sets containing both labels (e.g., $\{0, 1\}$) indicate uncertainty and are considered for deferral within the proposed CSP-CAD framework. This is showcased in the following section.

Table 3.4: Prediction set summary

| **Prediction Set** | **Interpretation** | **Action in CSP-CAD** |
|---|---|---|
| 0 | Confident normal prediction | Candidate for automated prediction |
| 1 | Confident failure prediction | Candidate for automated prediction |
| 0,1 | Ambiguous prediction | Candidate for deferral |

## 3.11 Confidence-Based Cost-Aware Deferral

Following probability calibration and conformal prediction, a confidence-based cost-aware deferral strategy was applied to determine whether an observation should receive an automated prediction or be deferred for further inspection. **Rather than requiring the classifier to make a prediction for every observation, the framework selectively abstains from predictions with insufficient confidence**. This approach reduces the likelihood of costly misclassifications in safety-critical predictive maintenance applications.

The deferral policy is based on calibrated prediction confidence and a cost-sensitive decision threshold that balances predictive coverage against operational risk.

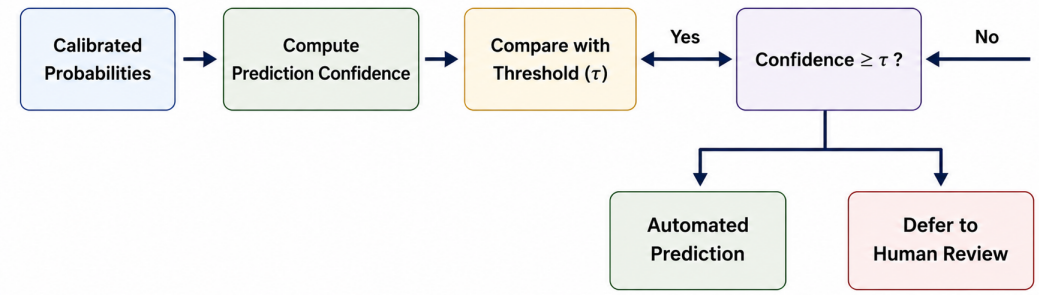



Figure 3.2: Confidence-Based Cost-Aware Deferral Framework

### 3.11.1 Deferral Procedure

For each test observation, prediction confidence was computed as the maximum calibrated class probability. This confidence value represents the classifier's certainty in its predicted class.

The confidence score was compared against a decision threshold, $\tau$. Observations with confidence greater than or equal to the threshold received an automated prediction, whereas observations with confidence below the threshold were deferred for additional inspection.

The threshold was selected using the calibration subset by minimizing an expected cost function that considers the relative costs of false negatives, false positives, and deferred decisions.

The **prediction confidence** is expressed as: $c(x) = \max_{y \in Y} \hat{p}(y|x)$ ..... (7) , where $c(x)$is the prediction confidence and $\hat{p}(y|x)$is the calibrated probability.

The **deferral decision rule** is expressed as:

$$d(x) = \begin{cases} \text{Predict}, & c(x) \geq \tau, \\ \text{Defer}, & c(x) < \tau. \end{cases} \quad \ldots\ldots (8)$$

where, $d(x)$ denotes the decision function for the input $x$, and $\tau$ represents the optimal confidence threshold calibrated to minimize the total operational expenditure associated with misclassification risks and expert intervention costs.

The expected cost is expressed as: $Cost = C_{FN}N_{FN}+C_{FP}N_{FP}+C_{DEF}N_{DEF}$ ..... (9)

where,$C_{FN}$, $C_{FP}$, and $C_{DEF}$ denote the respective costs of false negatives, false positives, and deferral interventions, while $N_{FN}$, $N_{FP}$, and $N_{DEF}$ represent the frequency of these outcomes within the calibration set.

The table 3.9 below summarizes the Decision Outcomes as a result of this workflow:

Table 3.5: Decision Interpretation

| **Confidence** | **Decision** | **Interpretation** |
|---|---|---|
| $c(x) \geq \tau$ | Predict | High-confidence prediction |
| $c(x) < \tau$ | Defer | Uncertain prediction requiring additional inspection |

## 3.12 Conformal Selective Prediction with Cost-Aware Deferral (CSP-CAD) Framework

The primary methodological contribution of this research is the adaptation of the Conformal Selective Prediction with Cost-Aware Deferral framework, as proposed by Kwon and Kim [25], to the predictive maintenance domain.The framework integrates calibrated probability estimation, Split Conformal Prediction, and confidence-based cost-aware deferral into a unified decision-making process for predictive maintenance. Unlike conventional prediction systems that always produce a single class prediction, CSP-CAD

explicitly identifies uncertain observations and defers them for further inspection when sufficient prediction confidence cannot be established.

The framework combines two complementary sources of uncertainty information. First, calibrated probabilities provide reliable confidence estimates for each prediction. Second, Split Conformal Prediction generates prediction sets that quantify predictive uncertainty while maintaining finite-sample coverage guarantees. These two uncertainty measures are jointly evaluated before issuing the final decision.

### 3.12.1 Decision Strategy

The CSP-CAD framework applies a two-stage decision process to each test observation. First, the conformal prediction set is examined. If the prediction set contains more than one candidate class, the observation is considered uncertain and is deferred for further inspection. If the prediction set contains a single class label, the calibrated prediction confidence is compared with the optimized confidence threshold, $\tau$. Automated predictions are issued only when both conditions indicate sufficient confidence; otherwise, the observation is deferred.

This hierarchical strategy enables the framework to combine uncertainty information derived from both conformal prediction and calibrated probability estimates before making an operational decision. The decision logic in defining the rule is:

$$D(x) = \begin{cases} \text{Predict}, & |\Gamma(x)| = 1 \wedge c(x) \geq \tau, \\ \text{Defer}, & \text{otherwise.} \end{cases} \quad \ldots\ldots (10)$$

where, $D(x)$ is the final decision, $\Gamma(x)$ is the conformal prediction set, $|\Gamma(x)|$denotes the number of labels in the prediction set, $c(x)$ is the calibrated prediction confidence and $\tau$ is the optimised confidence threshold. A summary of the decision outcomes is expressed in the table below:

Table 3.6: Decision Outcomes

| **Prediction Set** | **Confidence** | **Final Decision** |
|---|---|---|
| {0} | High | Predict Normal |
| {1} | High | Predict Failure |
| {0} | Low | Defer |
| {1} | Low | Defer |
| {0,1} | Any | Defer |

The adpated CSP-CAD framework combines conformal prediction and confidence-based selective prediction to improve the reliability of automated maintenance decisions. Conformal prediction provides uncertainty information through prediction sets, while

calibrated probabilities quantify confidence in individual predictions. By requiring agreement between both uncertainty measures before issuing an automated prediction, the framework reduces the likelihood of unreliable predictions and supports safer decision-making in highly imbalanced predictive maintenance applications.

### 3.12.2 Deferral Performance Analysis

The effectiveness of the cost-aware deferral policy was evaluated using several complementary metrics that quantify the trade-off between automation and decision reliability.

Coverage measures the proportion of observations for which an automated prediction is issued:

$$\text{Coverage} = \frac{\text{Retained Predictions}}{\text{Total Predictions}} \qquad \ldots\ldots (11)$$

Deferred Fraction measures the proportion of observations referred for additional inspection:

$$\text{Deferred Fraction} = \frac{\text{Deferred Predictions}}{\text{Total Predictions}} \qquad \ldots\ldots (12)$$

Retained Error quantifies the prediction error among non-deferred observations and serves as a measure of decision risk [25].

To further characterise the behaviour of the framework under varying confidence thresholds, three diagnostic analyses were performed:

1. **Risk–Coverage Analysis**, which evaluates how retained prediction error changes as prediction coverage decreases.

2. **Cost–Coverage Analysis**, which evaluates the relationship between operational cost and prediction coverage.

3. **Error-vs-Deferral Analysis**, which measures how predictive error evolves as increasingly uncertain observations are deferred.

Although the framework employs repeated stratified cross-validation to aggregate performance metrics across fifty independent evaluation runs, we acknowledge the importance of the exchangeability assumption fundamental to conformal prediction. Given that industrial time-series datasets, such as the AI4I dataset, may exhibit temporal dependencies or distribution drift, this assumption can be challenged [27]. To mitigate potential violations, the framework relies on stratified sampling to preserve consistent class distributions across cross-validation folds, thereby maintaining approximate exchangeability within the local data windows used for calibration and evaluation. Consequently, the resulting analyses provide statistically robust estimates of the trade-offs between predictive performance, uncertainty, operational cost, and selective prediction behaviour [28, 29].

The cost-aware deferral framework described in this section forms the baseline selective prediction strategy used throughout this research. The following section extends

this approach by integrating conformal prediction uncertainty directly into the deferral process, resulting in the adapted Conformal-Aware Deferral Framework.

### 3.12.3 Relationship to the Baseline Deferral Framework

The proposed Conformal-Aware Deferral Framework can be viewed as an extension of the Cost-Aware Deferral Framework presented in the work by Kwon and Kim [25].

The baseline framework evaluates only the magnitude of prediction confidence, whereas the proposed framework additionally considers the structure of the conformal prediction set. Consequently, the conformal-aware strategy provides an additional layer of uncertainty assessment before automated decisions are accepted.

This modification is particularly important for observations located near class boundaries, where confidence estimates may appear acceptable despite substantial uncertainty regarding the true class label. By explicitly identifying ambiguous conformal prediction sets, the framework is able to isolate difficult maintenance cases that may otherwise contribute disproportionately to operational risk.

### 3.12.4 Evaluation of the Proposed Framework

The effectiveness of the proposed Conformal-Aware Deferral Framework was assessed through comparison with the baseline confidence-based deferral strategy.

Performance was evaluated across the repeated stratified cross-validation framework using:

- Mean Risk–Coverage Curves: illustrating the relationship between retained prediction error and prediction coverage.
- Mean Cost–Coverage Curves: illustrating the relationship between expected operational cost and prediction coverage.The derivation of the costs used for this evaluation ($c_{FN}$, $c_{FP}$, and $c_{DEF}$) is critical for interpreting these curves. These values are hypothetical, representing a generalized industrial setting—an assumption necessitated by the lack of standardized cost methodologies in this field, as recently highlighted by Jamshidi et al.
- Mean Error-vs-Deferral Curves: illustrating how retained error changes as increasing numbers of uncertain predictions are deferred.
- Mean Cost per Sample: the empirical coverage achieved by the conformal prediction sets.
- Mean Conformal Coverage: the empirical coverage achieved by the conformal prediction sets.

- Mean Prediction Set Size: the average number of labels contained within each conformal prediction set.

These analyses quantify the extent to which conformal uncertainty information improves selective prediction performance and operational decision quality. By incorporating both calibrated confidence estimates and conformal prediction sets into the deferral process, the proposed framework seeks to achieve safer and more trustworthy maintenance recommendations while preserving strong predictive performance.

The resulting methodology establishes a direct connection between uncertainty quantification and decision-making, thereby transforming conformal prediction from a passive uncertainty estimation tool into an active component of maintenance decision support.

### 3.12.5 Comparative Analysis

Comparative experiments were conducted across the three machine learning models—Random Forest, RUSBoost, and XGBoost—to determine the robustness and generalizability of the proposed framework. **The results of the Confidence-Only Deferral policy were compared against those of the adapted Conformal-Aware Deferral policy to evaluate whether incorporating conformal uncertainty information leads to improved operational decision-making under class imbalance and uncertainty.** Additionally, the Area Under the Accuracy-Rejection Curve is calculated to quantify how effectively the framework's uncertainty estimates align with true prediction difficulty, confirming that higher performance gains are achieved by abstaining from the most ambiguous samples [30]. This evaluation approach emphasises the trade-off between risk and the degree of automation, enabling a business decision maker to identify a Pareto-optimal operating point for specific maintenance requirements.

The combination of predictive, calibration, conformal, and selective prediction evaluations provides a holistic assessment of model trustworthiness and practical deployment readiness for predictive maintenance applications.

# Chapter 4

# Analysis and Results

This section provides a rigorous examination of the experimental findings, focusing on the trade-offs between automated diagnostic accuracy and the reliability of deferred decisions under varying operational loads.

## 4.1 Baseline Classifier Evaluation

Before evaluating the proposed CSP-CAD framework using advanced ensemble learners, two classical machine learning algorithms—Logistic Regression and Decision Tree—were introduced as benchmark models. In addition, a Majority Class classifier was included as a trivial baseline. The objective of these baselines is to establish whether the proposed ensemble methods genuinely learn meaningful machine-failure patterns beyond what can be achieved through simple linear models, shallow decision boundaries, or naive majority-class prediction.

Because the AI4I dataset is highly imbalanced, with machine failures representing approximately 3.39% of observations, standard accuracy alone can produce misleading conclusions. Consequently, AUROC, AUPRC, Error Rate, and Recall were evaluated across 50 repeated stratified cross-validation runs.

## 4.2 Repeated Stratified CV: AUROC Comparison

Referencing Fig. 4.1, the Majority Class baseline achieved an AUROC of exactly 0.50, representing random guessing. This confirms that AUROC correctly identifies the absence of discriminative capability when all observations are assigned to the majority class. Logistic Regression achieved an AUROC of approximately 0.893, indicating that a linear decision boundary can capture a significant proportion of the machine-failure signal contained within the operational variables.

The Decision Tree further improved discrimination, achieving an AUROC of 0.908.

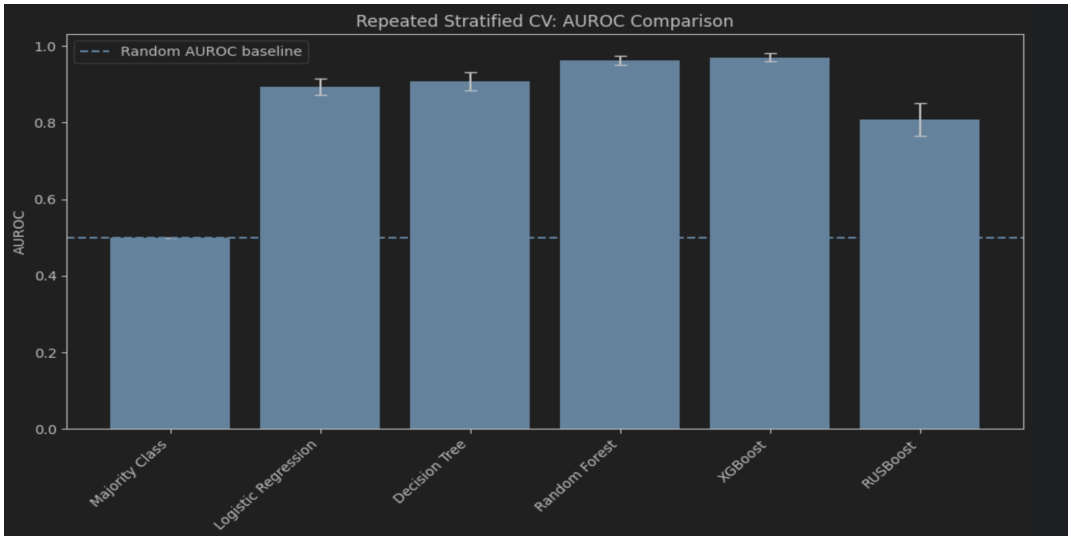



Figure 4.1: AUROC Results: These demonstrate substantial differences in discriminative ability among the evaluated models.

This suggests that non-linear relationships exist between machine operating conditions and failure occurrence. The ensemble methods substantially outperformed both baseline classifiers. Random Forest achieved an AUROC of 0.962, while XGBoost produced the highest AUROC of 0.970. These results indicate that ensemble-based learning methods are considerably more effective at ranking failure and non-failure observations than simpler baseline approaches. The relatively small standard deviations observed for Random Forest ($\pm 0.013$) and XGBoost ($\pm 0.011$) further demonstrate the stability of these models across repeated cross-validation folds.Key Finding

XGBoost exhibited the strongest overall ranking capability, while Random Forest achieved nearly equivalent performance. Both models substantially exceeded the performance of the baseline classifiers and the random predictor.

## 4.3 Repeated Stratified CV: AUPRC Comparison

While AUROC measures ranking ability, the Area Under the Precision-Recall Curve (AUPRC) is often more informative for highly imbalanced datasets because it focuses specifically on performance for the minority class. All machine learning models substantially exceeded the prevalence baseline, indicating a meaningful predictive signal within the dataset. [31]. The prevalence baseline of approximately 0.034 corresponds to the expected AUPRC of a random classifier and is represented by the dashed horizontal line.

Among the baseline models, Decision Tree outperformed Logistic Regression, suggesting that non-linear interactions between machine operating variables play an important role in failure prediction. Analysing Fig 4.2. the strongest performance was again observed for XGBoost (0.739) and Random Forest (0.721), both of which produced AUPRC

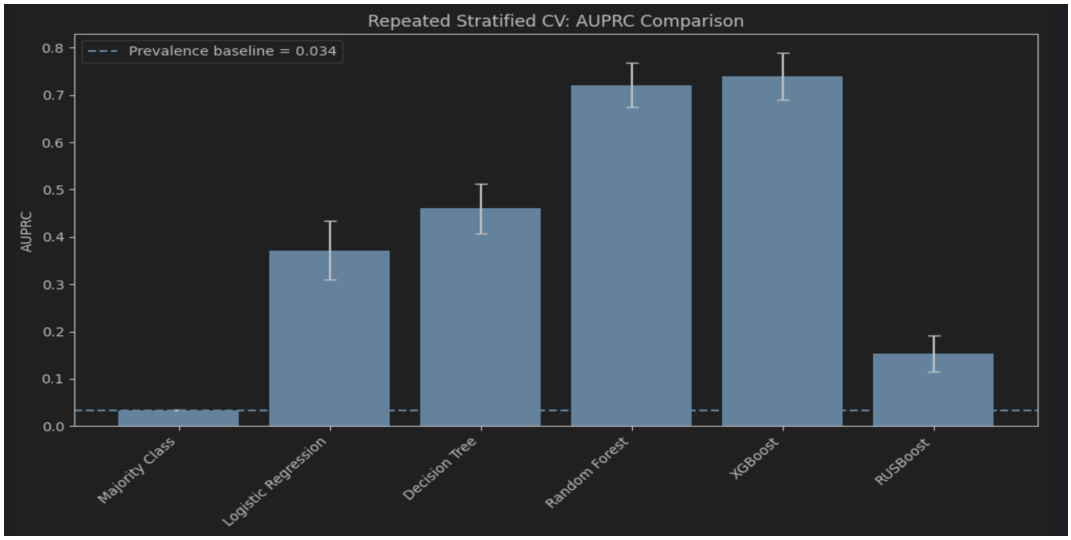



Figure 4.2: AUPRC Comparison of Repeated Stratified Results

values more than twenty times greater than the random baseline.

An important observation is that RUSBoost performed considerably worse than expected, achieving an AUPRC of only 0.153 despite being specifically designed for imbalanced classification. This suggests that, for the AI4I dataset, aggressive resampling may have reduced the model's ability to produce precise failure predictions.Key Finding

The AUPRC results confirm that Random Forest and XGBoost are substantially better at identifying true failure events while controlling false alarms. Because predictive maintenance is fundamentally concerned with detecting rare failures, AUPRC provides strong evidence that these two models are the most suitable candidates for the subsequent uncertainty-aware framework.

## 4.4 Comparative Analysis of Baseline Models

Table 4.1: Comparative Analysis of Baseline Models

| **model** | **AUROC** | **AUPRC** | **Recall** |
|---|---|---|---|
| Decision Tree | 0.9082 ± 0.0239 | 0.4603 ± 0.0521 | 0.2513 ± 0.1518 |
| Logistic Regression | 0.8928 ± 0.0217 | 0.3716 ± 0.0616 | 0.1964 ± 0.0932 |
| Majority Class | 0.5 ± 0.0 | 0.0339 ± 0.0002 | 0.0 ± 0.0 |
| RUSBoost | 0.8083 ± 0.0432 | 0.1528 ± 0.0385 | 0.0475 ± 0.0731 |
| Random Forest | 0.9622 ± 0.0126 | 0.7209 ± 0.0466 | 0.6041 ± 0.0803 |
| XGBoost | 0.9702 ± 0.011 | 0.7392 ± 0.0501 | 0.6478 ± 0.0894 |

The repeated cross-validation results in Table 4.1 reveal three important findings:

**Finding 1: Ensemble Models Dominate Classical Baselines**

Both Random Forest and XGBoost consistently achieved:

- Lower error rates
- Higher AUROC
- Higher AUPRC
- Higher recall

than Logistic Regression and Decision Tree. This indicates that ensemble learning provides substantial advantages for predictive maintenance applications involving complex interactions among machine operating variables.

**Finding 2: Class Imbalance Requires More Than Recall**

Although RUSBoost was explicitly designed for imbalanced learning, it achieved the lowest recall among the machine learning models (0.047).

This result demonstrates that imbalance-handling techniques alone do not guarantee strong predictive performance. The ability to learn meaningful discriminative structures appears more important than simply altering class distributions during training.

**Finding 3: XGBoost Emerges as the Strongest Base Classifier**

XGBoost achieved:

- Highest AUROC (0.970)
- Highest AUPRC (0.739)
- Highest Recall (0.648)
- Lowest Error Rate (0.018)

across the repeated cross-validation framework. These findings suggest that XGBoost provides the strongest predictive foundation for uncertainty quantification, conformal prediction, and cost-aware selective prediction.

## 4.5 Statistical Significance Analysis of Model Performance

To determine whether the observed differences in predictive performance between the evaluated machine learning models were statistically meaningful, statistical significance

testing was performed on the results obtained from the 50 repeated stratified cross-validation runs. Since each model was evaluated on identical train/test partitions across all repetitions, paired statistical tests were employed to preserve the dependency between observations. Following good statistical practice, the analysis consisted of:

1. **Friedman omnibus test** to determine whether any overall differences existed among the three models.
2. **Pairwise Wilcoxon signed-rank tests** with Holm correction to identify which models differed significantly.
3. **Paired t-tests** as a parametric sensitivity analysis.
4. **Effect size estimation (Cohen's $d_z$ and Rank-Biserial Correlation)** to quantify the magnitude of the observed differences.
5. **95% confidence intervals** for the paired mean differences to evaluate the precision of the estimated performance differences.

This hierarchical analysis provides stronger evidence than comparing mean AUROC values alone.

### 4.5.1 Friedman Omnibus Test

The Friedman test confirmed the presence of statistically significant disparities among the evaluated ensemble models, with a calculated chi-square statistic indicating that the performance rankings were unlikely to have occurred by chance. For AUROC, the Friedman test produced:

- $\chi^2 = 84.00$
- $p = 5.75 \times 10^{-5}$
- Kendall's W = 0.84

The Holm-adjusted p-value remained highly significant.

This result provides extremely strong statistical evidence that the three classifiers do not exhibit equivalent predictive performance across the 50 repeated cross-validation runs. The extremely small p-value indicates that the probability of observing differences of this magnitude under the null hypothesis of equal model performance is effectively zero.

Equally important is the value of **Kendall's W = 0.84**, which represents a very large effect according to commonly accepted interpretations. This indicates that the ranking of the models remained highly consistent across the repeated cross-validation experiments.

In other words, the superior model was not simply outperforming its competitors on a few folds but demonstrated consistently better performance throughout almost all repetitions.

This finding justifies proceeding to pairwise statistical comparisons to determine where these differences occurred.

### 4.5.2 Pairwise Wilcoxon Signed-Rank Tests

After establishing that significant overall differences existed, pairwise comparisons were performed using the Wilcoxon signed-rank test. This non-parametric test is preferable because repeated cross-validation score differences are not guaranteed to be normally distributed.

- RUSBoost vs XGBoost

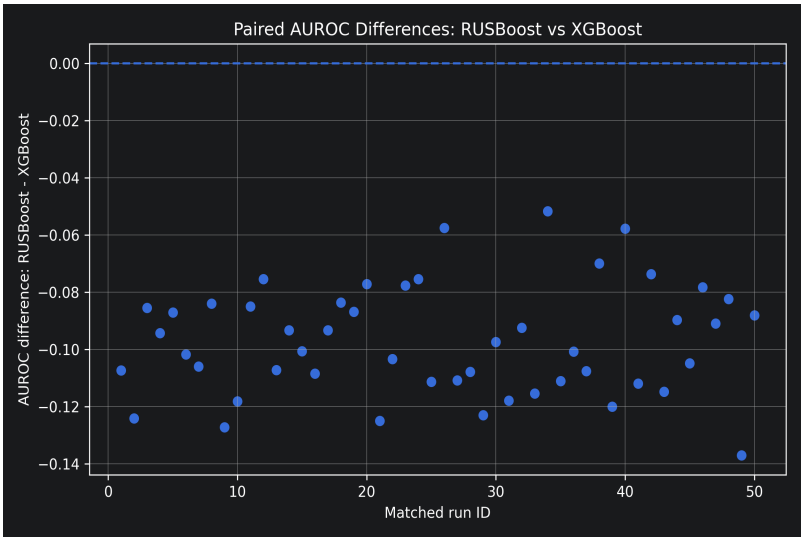



Figure 4.3: Pairwise Wilcoxon Signed-Ranked Test of RUSBoost vs XGBoost

This comparison of RUSBoost vs XGBoost in Fig 4.3 showed the largest performance difference.

Mean AUROC: RUSBoost = **0.86895** and XGBoost = **0.96599**

Mean paired difference: **0.0970**,

95% CI: [**0.1025, 0.0916**],

Holm-adjusted Wilcoxon p-value: $\mathbf{5.33 \times 10^{1}}$,

Rank-Biserial Correlation: −1.00

The Wilcoxon test indicates an overwhelmingly significant difference between RUSBoost and XGBoost. The confidence interval lies entirely below zero, confirming that XGBoost consistently achieved higher AUROC values across every repeated cross-validation run. The rank-biserial correlation of −1.00 represents the strongest possible effect, indicating that XGBoost outperformed RUSBoost in essentially every paired comparison.

This demonstrates that the observed superiority of XGBoost was systematic rather than being driven by a few exceptional folds.

- RUSBoost vs Random Forest

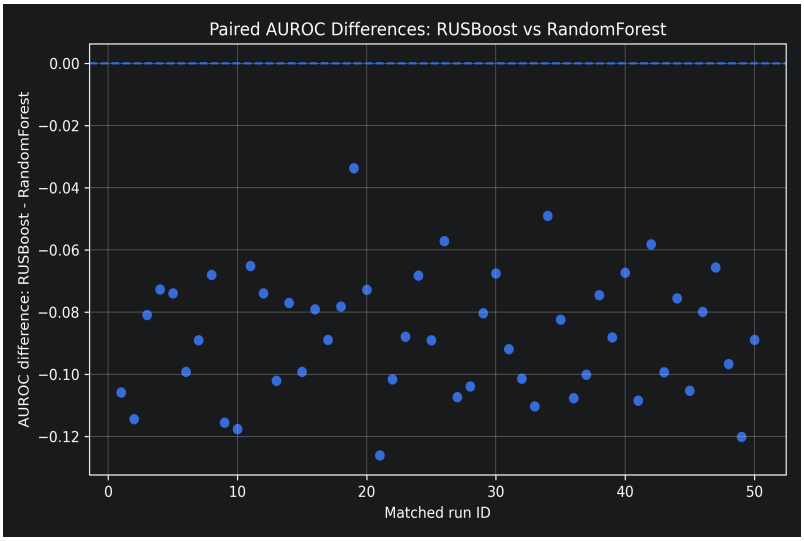



Figure 4.4: Pairwise Wilcoxon Signed-Ranked Test of RUSBoost vs Random Forest

This comparison of RUSBoost vs Random Forest in Fig 4.4 showed the largest performance difference.

Mean AUROC: RUSBoost = **0.86895** and Random Forest = **0.95627**

Difference: **0.0873**

95% CI: [**0.0929, 0.0817**],

Holm-adjusted p-value: $\mathbf{5.33 \times 10^{1}}$,

Rank-Biserial Correlation: −1.00

Random Forest also significantly outperformed RUSBoost across all repeated cross-validation runs. Similar to the previous comparison, the confidence interval excludes zero entirely, while the maximum possible rank-biserial correlation indicates extremely consistent superiority.

This confirms that the ensemble tree methods substantially improved discrimination performance relative to RUSBoost.

- Random Forest vs XGBoost

This comparison of Random Forest vs XGBoost in Fig 4.5 showed the largest performance difference. Mean AUROC: Random Forest = **0.95627** and XGBoost = **0.96599**

Difference: **0.00972**,

95% CI: [**0.01363, 0.00582**],

Holm-adjusted p-value: **1.98 × 10**,

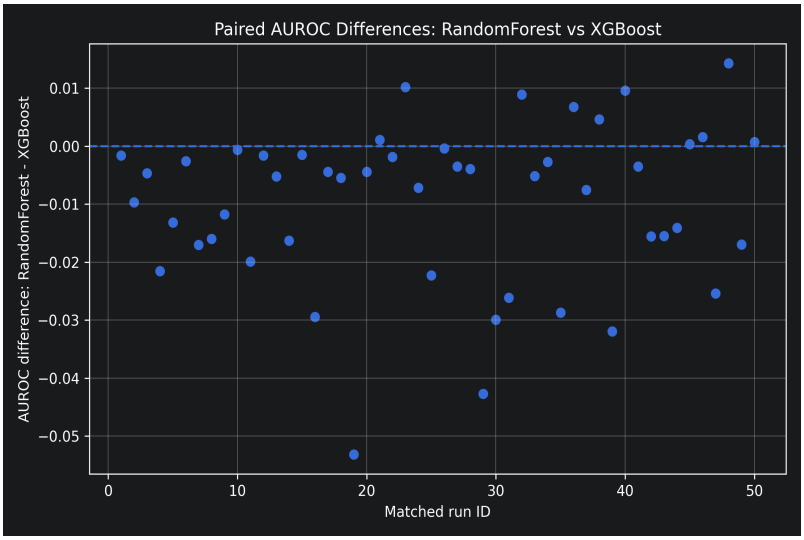



Figure 4.5: Pairwise Wilcoxon Signed-Ranked Test of Random Forest vs XGBoost

Rank-Biserial Correlation: $-0.725$

Although the absolute AUROC improvement was relatively small (approximately 0.01), the difference remained statistically significant after Holm correction. The confidence interval excludes zero, demonstrating that XGBoost consistently achieved slightly higher AUROC values than Random Forest across the repeated validation runs.

However, this result also illustrates the distinction between **statistical significance** and **practical significance**. Because repeated cross-validation substantially reduces estimation variance, even modest but consistent improvements can become statistically significant. Therefore, while XGBoost demonstrated superior discrimination performance, the practical improvement over Random Forest was considerably smaller than the improvements observed relative to RUSBoost.Interpretation of the Paired Difference Plots

The paired AUROC difference plots provide an intuitive visual confirmation of the statistical tests.RUSBoost vs XGBoost

Every point lies below the zero reference line, indicating that XGBoost achieved higher AUROC values in every matched repeated cross-validation run. Furthermore, the differences are relatively uniform, suggesting highly stable superiority rather than occasional improvements.RUSBoost vs Random Forest

A similar pattern is observed, with all paired differences remaining below zero. This demonstrates that Random Forest consistently outperformed RUSBoost throughout the repeated validation procedure.Random Forest vs XGBoost

The differences are much closer to zero and exhibit greater variability. Nevertheless, most observations remain below zero, indicating that XGBoost generally achieved slightly higher AUROC values than Random Forest. The smaller spread around zero is consistent with the more modest effect size observed in the statistical analysis.

### 4.5.3 Statistical Analysis Conclusion

The statistical significance analysis confirms that the observed differences in predictive performance are not attributable solely to random variation across the repeated cross-validation folds. The Friedman omnibus test demonstrated highly significant overall differences among the three classifiers, while subsequent Holm-adjusted Wilcoxon signed-rank tests identified significant pairwise differences between every model pair. Confidence intervals consistently excluded zero, further confirming the stability of the observed performance differences, and effect size analysis revealed that the improvements of XGBoost and Random Forest over RUSBoost were both statistically and practically substantial. Although XGBoost also significantly outperformed Random Forest, the absolute improvement in AUROC was relatively small, suggesting that its advantage is statistically reliable but operationally modest. Consequently, the statistical evidence supports the conclusion that **XGBoost provides the strongest overall discrimination performance, Random Forest represents a close second with competitive performance, and RUSBoost consistently exhibits inferior discriminative ability on this predictive maintenance task**. This interpretation is more rigorous than relying solely on mean AUROC values and provides formal statistical justification for the comparative conclusions presented in this thesis.

### 4.5.4 Calibration Performance Analysis

#### Mean Expected Calibration Error (ECE)

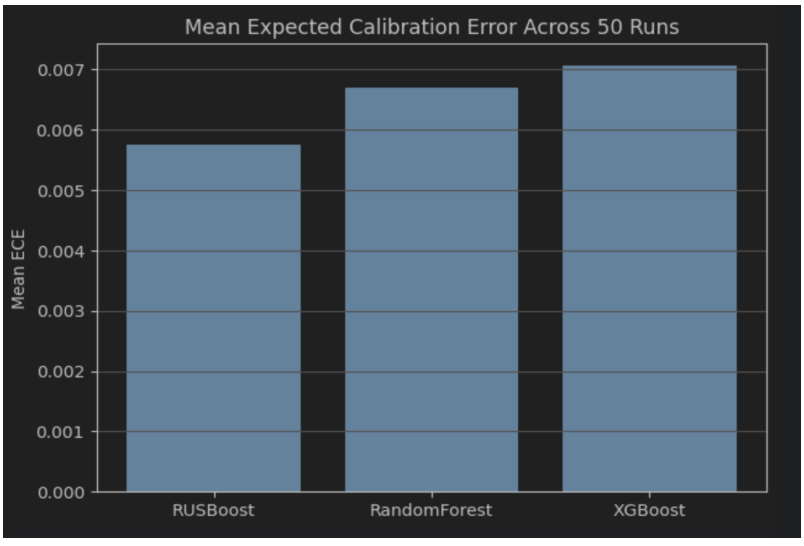



Figure 4.6: Plot of Mean Expected Calibration Error(ECE) Across 50 Runs

Probability calibration is a critical component of the proposed CSP-CAD framework because both conformal prediction and cost-aware deferral rely on reliable probability estimates. While a classifier may achieve high predictive accuracy, poorly calibrated

probabilities can lead to inappropriate confidence estimates and unreliable uncertainty quantification [19].

To evaluate calibration quality, the Expected Calibration Error (ECE) was computed across all fifty repeated stratified cross-validation runs. The Expected Calibration Error was evaluated using **10 equally spaced probability bins**. This choice provided stable calibration estimates across all 50 repeated cross-validation runs and is consistent with standard calibration evaluation practice. Consequently, the reported Mean ECE values represent average calibration performance over 50 repeated experiments rather than a single train-test split.The plot of the used model's ECE values is shown in Fig 4.6. ECE measures the average discrepancy between predicted confidence and observed outcome frequencies, where lower values indicate better calibration [8].

All three ensemble models achieved exceptionally low ECE values, indicating excellent probability calibration following isotonic regression. The differences between models are relatively small, suggesting that the calibration procedure successfully corrected most systematic probability distortions.

RUSBoost achieved the lowest ECE, indicating the closest alignment between predicted probabilities and observed frequencies. However, as demonstrated in the previous section, superior calibration alone does not necessarily translate into superior predictive performance.

Random Forest and XGBoost exhibited slightly higher ECE values but remained well below 0.01, indicating highly reliable probability estimates suitable for uncertainty-aware decision-making.Key Finding

The calibration results demonstrate that isotonic regression successfully transformed raw classifier outputs into reliable probability estimates. Consequently, the calibrated probabilities provide a trustworthy foundation for both conformal prediction and cost-aware selective prediction.

## 4.6 Conformal Prediction Analysis

### 4.6.1 Conformal Coverage Performance

The objective of conformal prediction is to provide statistically valid uncertainty estimates with finite-sample coverage guarantees [32]. In this research, a target coverage level of 90% ($\alpha = 0.10$) was specified. Conformal coverage measures the proportion of observations whose true labels are contained within the generated prediction sets [33].

The average conformal coverage achieved across fifty repeated cross-validation runs is shown in Fig. 4.7. All three models achieved empirical coverage levels slightly above the nominal 90% target. This result is important because it demonstrates that the conformal prediction framework maintained its theoretical validity across multiple independent

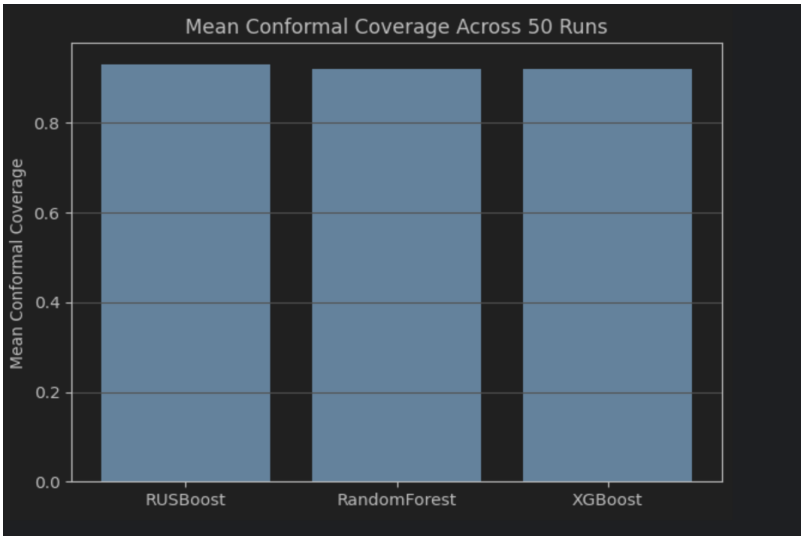



Figure 4.7: Plot of Mean Conformal Coverage Across 50 Runs

resamples of the dataset.

The slight conservatism observed in the coverage values is expected and desirable in safety-critical applications. Over-coverage indicates that prediction sets are marginally larger than required, reducing the likelihood of excluding the true machine state.

- **Key Finding**: The conformal prediction framework consistently achieved the desired coverage level across all evaluated models, validating the use of conformal uncertainty estimates within the proposed maintenance decision-support framework.

## 4.7 Cost-Aware Deferral Analysis

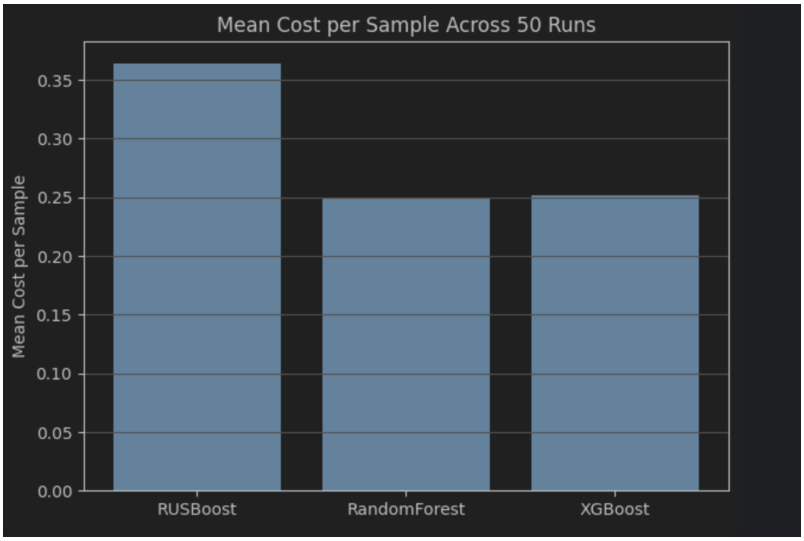



Figure 4.8: Plot of Mean Cost per Sample Across 50 Runs

### 4.7.1 Operational Cost Analysis

Predictive maintenance systems are ultimately deployed to support operational decision-making. Consequently, model evaluation should not be restricted to predictive performance alone but should also consider the economic consequences of incorrect decisions [34].

To assess operational effectiveness, the proposed framework evaluated the average cost incurred per observation under the cost-aware deferral policy. The plot of the results is shown in Fig 4.8. Random Forest achieved the lowest average operational cost, closely followed by XGBoost. Despite XGBoost producing the strongest predictive performance metrics, Random Forest achieved a marginally lower cost per sample. This suggests that Random Forest may produce a more favourable balance between false negatives, false positives, and deferred decisions under the chosen cost structure. RUSBoost incurred substantially higher operational costs than the other models. This is consistent with its weaker predictive performance and higher retained error rates observed throughout the evaluation.

- **Key Finding**: Random Forest and XGBoost provide substantially more cost-effective maintenance recommendations than RUSBoost, indicating that stronger predictive discrimination directly translates into improved operational outcomes.

## 4.8 Proposed Conformal-Aware Deferral Analysis

This section evaluates the central contribution of the thesis.

### 4.8.1 Risk-Coverage Analysis

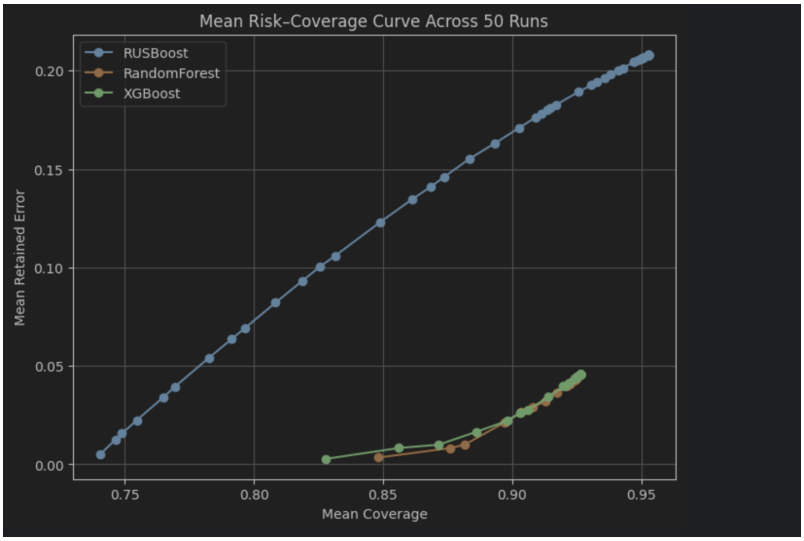



Figure 4.9: Plot of Mean Mean Risk-Coverage Curve Across 50 Runs

Risk-Coverage analysis evaluates how prediction risk changes as increasingly uncertain observations are removed from automated decision-making.

Coverage represents the proportion of observations receiving automated predictions, while risk corresponds to the retained prediction error among those predictions. Several observations emerge from Fig. 4.9.

Random Forest and XGBoost maintain extremely low retained error rates across most coverage levels. Even when operating at coverage values exceeding 90%, both models achieve retained error rates below 5%. In contrast, RUSBoost exhibits substantially higher risk across the entire coverage range. At comparable coverage levels, retained error is approximately four to five times higher than that observed for Random Forest and XGBoost. As coverage decreases, all models experience declining risk, confirming that the deferral policy successfully removes increasingly uncertain observations from automated decision-making.

- **Key Finding**: The proposed selective prediction framework effectively reduces prediction risk by deferring uncertain observations, with Random Forest and XGBoost benefiting most from uncertainty-aware decision-making.

### 4.8.2 Cost-Coverage Analysis

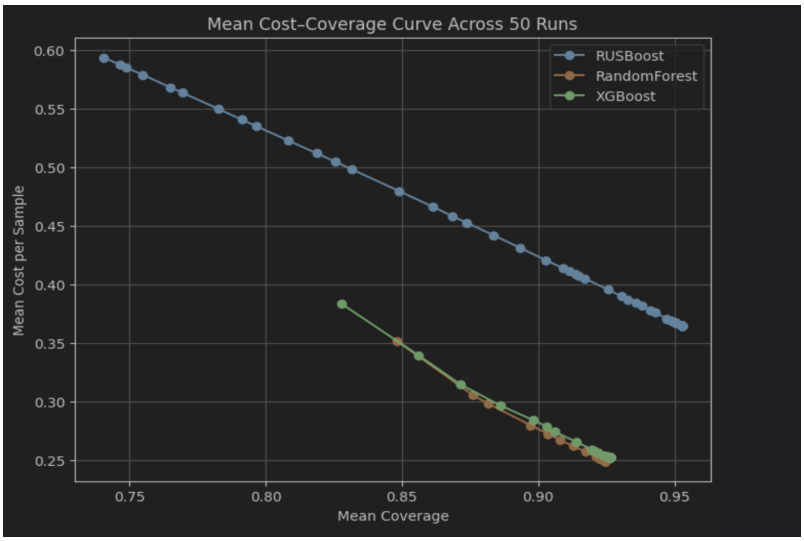



Figure 4.10: Plot of Mean Risk-Coverage Curve Across 50 Runs

Cost-Coverage analysis evaluates the relationship between operational cost and prediction coverage. The results indicate a clear inverse relationship between cost and coverage. As coverage decreases and more uncertain observations are deferred, operational cost also decreases.

Random Forest and XGBoost consistently achieve lower costs than RUSBoost throughout the evaluated coverage range.Several observations emerge from Fig. 4.10. For Random Forest and XGBoost, the lowest observed costs occur at coverage values around

92–93%, suggesting that modest levels of selective deferral can substantially improve operational efficiency. RUSBoost requires significantly lower coverage levels to achieve comparable cost reductions, indicating that a larger proportion of observations must be deferred before acceptable performance can be achieved.

- **Key Finding**: The cost-aware deferral framework successfully identifies operating regions where selective prediction reduces expected maintenance costs without requiring excessive levels of human intervention.

### 4.8.3 Error-vs-Deferral Analysis

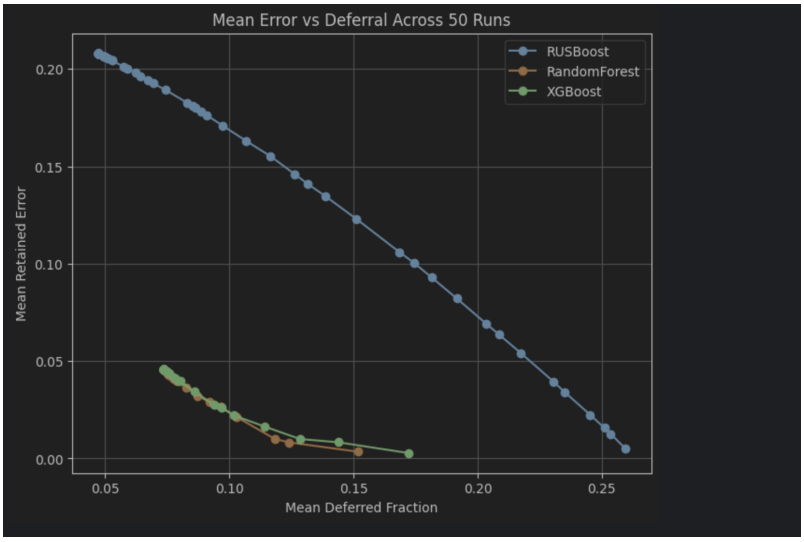



Figure 4.11: Plot of Mean Error vs. Deferral Curve Across 50 Runs

The Error-vs-Deferral curve directly evaluates the trade-off between automation and reliability. As the deferred fraction increases, retained error consistently decreases across all models.

This behaviour supports the rationale for selective prediction: models may abstain from making predictions on instances with high uncertainty, which often contribute disproportionately to overall error [22]. Several observations emerge from Fig. 4.11.

Random Forest and XGBoost demonstrate the most favourable trade-off. Small increases in deferral rates produce substantial reductions in retained error, indicating that these models effectively identify genuinely uncertain observations.

RUSBoost exhibits a less efficient trade-off, requiring significantly higher deferral rates before comparable error reductions are achieved.

- **Key Finding**: The selective prediction mechanism effectively improves prediction reliability by removing uncertain observations from automated decision-making, with Random Forest and XGBoost showing the strongest benefit.

# Chapter 5

# Conclusion and Results Discussion

This chapter summarizes the key findings of the research and discusses their implications for predictive maintenance and uncertainty-aware decision support systems. The limitations of the study are subsequently examined, followed by recommendations for future research directions that could further enhance the proposed Conformal Selective Prediction with Cost-Aware Deferral (CSP-CAD) framework.

## 5.1 Summary of Research Findings

The primary objective of this research was to investigate whether calibrated probabilistic machine learning models, conformal prediction, and cost-aware selective prediction could be integrated into a unified framework capable of producing safer and more reliable maintenance decisions in highly imbalanced industrial environments.

To achieve this objective, a Conformal Selective Prediction with Cost-Aware Deferral (CSP-CAD) framework was developed and evaluated using the AI4I 2020 Predictive Maintenance dataset. The framework combined machine learning-based failure prediction, isotonic probability calibration, split conformal prediction, cost-aware threshold optimization, and selective prediction mechanisms within a repeated stratified cross-validation evaluation framework consisting of fifty independent experimental runs.

The experimental results demonstrated that ensemble learning approaches substantially outperformed traditional baseline models. XGBoost achieved the strongest predictive performance, obtaining the highest AUROC and AUPRC values among all evaluated classifiers, while Random Forest achieved comparable predictive performance and the lowest average operational cost.

Probability calibration results indicated that isotonic regression successfully produced highly reliable probability estimates, with all evaluated models achieving Expected Calibration Error (ECE) values below 0.01. These results confirmed that calibrated probabilities provide a robust foundation for uncertainty-aware decision-making.

The conformal prediction framework consistently achieved empirical coverage levels close to the target 90% confidence level across all models. This finding validated the use of conformal prediction as a statistically reliable uncertainty quantification mechanism for predictive maintenance applications.

The selective prediction experiments further demonstrated that uncertainty-aware deferral can substantially improve decision quality. Risk-Coverage, Cost-Coverage, and Error-vs-Deferral analyses showed that selectively deferring uncertain observations reduced prediction error and operational cost while maintaining high levels of automation.

Collectively, these findings provide strong evidence that integrating calibrated probabilities, conformal prediction, and cost-aware selective prediction can improve both the reliability and operational usefulness of predictive maintenance systems.

## 5.2 Achievement of Research Objectives

This research successfully achieved its stated objectives.

**Objective 1:**

**To develop predictive machine learning models capable of identifying machine failures in a highly imbalanced industrial dataset.**

This objective was achieved through the development and evaluation of Logistic Regression, Decision Tree, RUSBoost, Random Forest, and XGBoost classifiers. Ensemble learning models consistently achieved superior predictive performance, demonstrating the feasibility of accurate machine failure prediction using operational sensor data.

**Objective 2:**

**To improve the reliability of model confidence estimates through probability calibration.**

This objective was achieved through the application of isotonic regression calibration. Experimental results demonstrated extremely low calibration errors across all evaluated models, indicating strong agreement between predicted probabilities and observed outcomes.

**Objective 3:**

**To quantify predictive uncertainty using conformal prediction.**

This objective was achieved through the implementation of Split Conformal Prediction. Empirical coverage results confirmed that the conformal framework consistently satisfied the desired uncertainty guarantees.

**Objective 4:**

**To develop a cost-aware selective prediction framework capable of reducing operational risk.**

This objective was achieved through the introduction of a cost-based deferral mechanism that balances false-negative costs, false-positive costs, and human-review costs. The resulting framework successfully reduced expected operational costs across multiple decision thresholds.

**Objective 5:**

**To investigate the integration of conformal prediction into selective prediction decisions.**

This objective represents the primary contribution of the thesis. The proposed Conformal-Aware Deferral Framework successfully incorporated conformal prediction sets into the decision-making process, enabling uncertain observations to be automatically identified and referred for expert review. Experimental results demonstrated that uncertainty-aware deferral can improve both prediction reliability and operational decision quality.

## 5.3 Research Contributions

The major contributions of this research are summarized as follows:

**Contribution 1: Development of the CSP-CAD Framework**

This research proposed a novel Conformal Selective Prediction with Cost-Aware Deferral (CSP-CAD) framework that integrates predictive modelling, probability calibration, conformal prediction, and selective prediction within a unified maintenance decision-support pipeline.

**Contribution 2: Integration of Conformal Prediction into Deferral Decisions**

Unlike conventional selective prediction systems that rely solely on confidence scores, the proposed framework incorporates conformal prediction sets directly into the deferral decision process. This allows uncertainty estimates to actively influence maintenance recommendations.

**Contribution 3: Evaluation Using Repeated Stratified Cross-Validation**

The framework was evaluated using a repeated stratified cross-validation methodology consisting of fifty independent experimental runs. This provides a substantially more

robust estimate of performance than conventional single train-test evaluations commonly reported in predictive maintenance literature.

**Contribution 4: Operational Evaluation Beyond Classification Metrics**

The research extends conventional predictive maintenance evaluation by introducing operationally meaningful analyses, including:

- Risk-Coverage analysis
- Cost-Coverage analysis
- Error-vs-Deferral analysis
- Mean Cost per Sample evaluation

These metrics directly assess the practical value of predictive maintenance systems in real-world decision-making environments.

## 5.4 Limitations of the Research

Despite the promising results achieved by the proposed framework, several limitations should be acknowledged.

### 5.4.1 Dataset Limitation

The research was conducted using the AI4I 2020 Predictive Maintenance dataset, which is a publicly available benchmark dataset [35]. Although widely used in predictive maintenance research, the dataset represents a simulated industrial environment and may not fully capture the complexity, noise characteristics, and operational variability encountered in real industrial systems [36].

### 5.4.2 Binary Failure Formulation

The study focused on binary machine failure prediction rather than multi-failure diagnosis. While this simplification facilitates uncertainty quantification and selective prediction analysis, industrial maintenance systems often require identification of specific failure modes and root causes.

### 5.4.3 Exchangeability Assumption

The validity guarantees provided by conformal prediction rely on the assumption that calibration and future observations are exchangeable. In real-world industrial environments, sensor drift, equipment aging, changing operating conditions, and concept drift may violate this assumption and potentially affect coverage performance [37].

### 5.4.4 Cost Function Simplification

The cost model employed in this research uses predefined penalty values for false negatives, false positives, and deferred decisions. In practice, maintenance costs may vary across assets, operating conditions, production schedules, and safety requirements. Consequently, the selected cost structure may not fully represent all industrial deployment scenarios.

### 5.4.5 Human Expert Assumption

Deferred observations were assumed to be reviewed by human experts capable of making correct decisions. However, human decision-makers are also subject to uncertainty, variability, and potential errors. The framework therefore evaluates the effectiveness of deferral rather than the complete human-in-the-loop maintenance process.

### 5.4.6 Limited Model Diversity

Although several machine learning models were evaluated, the study did not investigate deep learning architectures, Bayesian neural networks, transformer-based approaches, or advanced uncertainty-aware learning methods. These approaches may provide additional improvements in predictive performance and uncertainty estimation.

## 5.5 Recommendations and Future Work

Several opportunities exist to extend and enhance the proposed framework.

### 5.5.1 Evaluation on Real Industrial Datasets

Future research should validate the CSP-CAD framework using operational maintenance datasets collected from real industrial environments such as manufacturing systems, energy infrastructure, transportation systems, and offshore oil and gas facilities. This would provide stronger evidence regarding the framework's practical applicability.

### 5.5.2 Multi-Class Failure Diagnosis

Future work should extend the framework from binary failure prediction to multi-class fault diagnosis. This would enable the system not only to predict whether a failure is likely to occur but also to identify the specific failure mechanism requiring maintenance intervention.

The current implementation uses standard Split Conformal Prediction. Future research could investigate adaptive conformal prediction techniques, including class-conditional

conformal prediction, Mondrian conformal prediction, and online conformal prediction methods, which may improve coverage performance under changing operating conditions.

### 5.5.4 Dynamic Cost Modelling

Future work should explore adaptive cost functions that incorporate asset criticality, maintenance schedules, production losses, and safety consequences. Such models could provide more realistic decision support for industrial maintenance planning.

### 5.5.5 Human-in-the-Loop Evaluation

The proposed framework assumes that deferred observations are reviewed by human experts. Future research should evaluate the interaction between machine learning systems and maintenance engineers to better understand how uncertainty information influences real-world maintenance decisions. Drawing on findings by De Toni et al., which suggest that conformal prediction sets may not be the optimal presentation format for human experts compared to other uncertainty representations, future work should explicitly examine whether the currently proposed conformal sets actively improve, or perhaps hinder, human decision-making accuracy. Furthermore, integrating causal anomaly detection methods may enhance system robustness by reducing false alarms triggered by non-anomalous out-of-distribution data.

### 5.5.6 Explainable Deferral Decisions

Future research could integrate explainable artificial intelligence (XAI) techniques such as SHAP or LIME with the proposed framework. This would allow maintenance engineers to understand not only why a failure prediction was generated but also why a particular observation was deferred.

### 5.5.7 Bayesian and Deep Learning Extensions

Future work could investigate Bayesian neural networks, deep ensembles, evidential learning, and probabilistic deep learning approaches. These methods may provide richer uncertainty estimates and potentially improve selective prediction performance.

### 5.5.8 Deployment-Oriented Evaluation

Finally, future research should investigate real-time deployment of the framework within industrial monitoring systems. This would allow evaluation of computational efficiency, scalability, and long-term operational benefits under realistic maintenance conditions. Regarding user acceptance, it is critical to engage with recent findings on human-AI complementarity (e.g., De Toni et al., 2024), which suggest that standard conformal

sets may not be the optimal presentation format for human experts in high-stakes decisions; exploring alternative uncertainty representations and their interaction with human decision-makers remains a critical aspect for real-world deployment.

### 5.5.9 Comparison with Established Learning to Defer (L2D) Strategies

While comparison against confidence-based thresholding is important, it is a limited baseline. The field of 'Learning to Defer' has developed sophisticated approaches to jointly train predictors and rejectors. Additionally, recent work like 'No Need for Learning to Defer? A Training Free Deferral Framework to Multiple Experts through Conformal Prediction' (Bary et al., 2025) presents a direct, training-free CP-based competitor. Including these as benchmarks would significantly strengthen the claim of superiority. [38]

## 5.6 Conclusion

This thesis presented a Conformal Selective Prediction with Cost-Aware Deferral framework for predictive maintenance in highly imbalanced industrial environments, which builds upon the methodological foundation recently established for clinical triage applications by Kwon and Kim [25]. The framework integrates machine learning-based failure prediction, isotonic probability calibration, split conformal prediction, cost-aware threshold optimisation, and selective prediction within a unified decision-support architecture.

Experimental evaluation using repeated stratified cross-validation demonstrated that ensemble learning models, particularly XGBoost and Random Forest, provide strong predictive performance while maintaining highly reliable probability estimates. The conformal prediction component successfully delivered statistically valid uncertainty estimates, achieving empirical coverage levels close to the desired confidence threshold.

Most importantly, the proposed Conformal-Aware Deferral Framework showed that uncertainty information can be incorporated directly into maintenance decision-making. By selectively deferring uncertain observations, the framework reduced prediction risk and operational cost while preserving high levels of automated decision support.

The findings demonstrate that predictive maintenance systems should not be evaluated solely on predictive accuracy. Instead, reliable uncertainty quantification and selective decision-making mechanisms are essential for developing trustworthy artificial intelligence systems capable of supporting high-stakes industrial operations. Consequently, the proposed CSP-CAD framework represents a meaningful step toward safer, more transparent, and economically informed predictive maintenance decision-support systems.

## 5.7 Key Concepts and Terminology

To facilitate understanding of the technical terminology used in this study, the following key concepts are defined:

- **Predictive Maintenance**: A strategy that monitors asset condition to optimise maintenance scheduling [39].
- **Conformal Prediction**: A statistical framework that produces uncertainty sets with guaranteed coverage probabilities [32]. Note that this framework relies on the assumption of exchangeability; because industrial time-series data often exhibit temporal dependencies and distribution shift, this assumption is frequently violated, necessitating specialized adaptations to maintain valid coverage.
- **Selective Prediction**: A machine learning paradigm in which the model may abstain from making a prediction when uncertainty is high [40].
- **Cost-Aware Deferral**: A decision mechanism that incorporates the economic consequences of misclassification, such as false positives or false negatives, into the threshold optimisation process [25].
- **Isotonic Calibration**: A post-processing technique used to transform raw model scores into well-calibrated, reliable probability estimates [41].

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